STRUCTURAL LIVENESS OF CONSERVATIVE PETRI NETS: DOUBLY EXPONENTIAL LIVE MARKINGS AND EXPSPACE COMPLETENESS

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Editorial scope. One conservative/structurally bounded Petri structural-liveness complexity group, double-exponential live-marking upper bound and ordinary reversible PP-net EXPSPACE lower bound together. General Petri-net structural-liveness complexity remains explicitly open. Minimal-linear-system bounds, special subclasses and direct consequences are not separate targets.

Allowed standard prerequisites (editorial).

- Author-cited standard foundations. All local definitions, main supporting constructions and complete source bibliography are retained; cited standard foundations remain original references, with no new proof or assumption.

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Original abstract. We show that the EXPSPACE-hardness result for structural liveness of Petri nets [Jančar and Purser, 2019] holds even for a simple subclass of conservative nets. As our main result, we prove that for structurally live conservative nets, the values of the minimal live markings are at most doubly exponential in the size of the net. This implies the EXPSPACE-completeness of structural liveness for conservative Petri nets. The result also applies to structurally bounded Petri nets, whereas the complexity of the general case remains open.

As a proof ingredient of independent interest, we present an extension of known results on the bounds of minimal integer solutions to Boolean combinations of linear equalities, inequalities, and divisibility constraints.

1. Introduction

Petri nets are a well-known model of a class of distributed systems; for an introduction, we refer, *e.g.*, to the monographs [Rei13] and [BD24].

The *reachability* problem asks whether a given target marking is reachable from a given initial marking. Here, a *marking* is the standard term for a Petri net configuration, assigning a number of *tokens* to each *place* (local state). The reachability problem for Petri nets is well known for its computational complexity: its Ackermann-completeness has only recently been established (see [Ler21, CO21] for the lower bound and [LS19] for the upper bound).

Given a Petri net and an initial marking, the *boundedness* problem asks whether the reachability set is finite, and the *liveness* problem asks whether all *transitions* (system actions) remain live forever. Both are among the standard analysis problems. While boundedness is known to be EXPSPACE-complete [CLM76, Rac78], liveness is tightly related to reachability [Hac74] and is thus now known to be Ackermann-complete.

There are also natural structural versions of boundedness and liveness. The *structural boundedness* problem asks, given a Petri net, whether the net is bounded for every initial marking, while the *structural liveness* problem asks, given a Petri net, whether there exists an initial marking for which the net is live.

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While structural boundedness is easily shown to be in PTIME (and is thus substantially easier than boundedness), structural liveness is only known to be EXPSPACE-hard and decidable [JP19]. In fact, the decidability result can be strengthened by recent results on the home-space problem [JL23, JL24], which readily imply an Ackermannian upper bound for structural liveness as well; nevertheless, the large complexity gap still calls for clarification.

Our contribution. As a step towards clarifying the complexity of *structural liveness* for general Petri nets, we establish *EXPSPACE-completeness* for the class of *conservative nets*, which preserve a weighted sum of tokens throughout their executions. We recall that deciding whether a given net is conservative is also in PTIME, similarly to structural boundedness (see, *e.g.*, [MW14]).

A crucial notion in our proof is *structural reversibility*, referred to simply as *reversibility* in this paper. A net is reversible if there exists a sequence of transitions in which each transition occurs at least once and whose overall effect is zero, that is, executing the sequence does not change the marking. Equivalently, the effect of any transition can be "undone" by some transition sequence. Reversibility is also in PTIME, and it can be easily shown to be a necessary condition for structural liveness in the case of structurally bounded nets.

Moreover, it is immediate that every conservative net is structurally bounded, and a straightforward application of Farkas' lemma shows that every reversible structurally bounded net is conservative. Hence, our EXPSPACE-completeness result can equivalently be stated for the class of *structurally bounded nets*.

As a natural first step in our future research, we plan to address several subtle issues that would allow us to extend the result to the full class of reversible nets.

The lower bound, EXPSPACE-hardness, is achieved by adapting the construction of [JP19], which shows a reduction from the EXPSPACE-complete word problem for commutative semigroups. This problem can also be phrased as a coverability problem for reversible Petri nets [CLM76, MM82, May97]. We recall that coverability is a weaker form of reachability: it asks whether there exists a reachable marking that is component-wise at least as large as the target. Our adaptation demonstrates EXPSPACE-hardness of structural liveness even for nets where each transition has precisely two input and two output places, that is, for nets that naturally correspond to population protocols [AAD$^+$06].

Our main result is an EXPSPACE upper bound. The crucial step establishes that for every structurally live conservative net, there exists a live marking with at most a doubly exponential (2-exp) number of tokens. In fact, we establish this 2-exp bound in a more general setting: for nets that possess live bottom strongly connected components in their reachability graphs. Conservative nets constitute a special case, for which such a 2-exp upper bound readily implies an EXPSPACE upper bound for the structural liveness problem.

The aforementioned 2-exp bound is achieved by showing that for any such net, there exists a specific quantifier-free Presburger formula that admits a "small" (2-exp) solution and whose solutions are live markings. More precisely, a solution to this formula represents a set of at most exponentially many mutually reachable markings, which together demonstrate that all of them are live.

A key aspect of our approach is the replacement of reachability with a conceptually simpler *virtual reachability*, which allows for negative numbers of tokens in (virtual) markings. To express virtual reachability in reversible nets, we employ linear systems—that is, Boolean combinations of linear equalities, inequalities, and divisibility constraints—which constitute the aforementioned special Presburger formulas. As a technical contribution of independent interest, we provide exponential bounds on the minimal solutions of such linear systems, extending known results such as those in [Pot91] (see also the survey [Haa18]).

Related research. This paper can be viewed as a continuation of the research line initiated by [BE16], which explicitly indicated that even the decidability of structural liveness for Petri nets remained open at that time. As already mentioned, decidability and EXPSPACE-hardness are now established [JP19]; furthermore, the decidability result was recently strengthened by [JL23, JL24], implying an Ackermannian upper bound. Another result in this research line is the PSPACE-completeness of structural liveness for IO-nets (Immediate Observation Petri Nets), which were introduced in [ERW19] and inspired

by a subclass of population protocols in [AAER07]. While [ERW19] does not consider structural liveness explicitly, a PSPACE upper bound follows directly from its results on liveness; an explicit, self-contained proof of PSPACE-completeness is provided in [JV23].

Numerous studies have investigated liveness for various subclasses of Petri nets, often exploring related structural properties (see the monographs [Rei13, BD24, DE95], or [HD18] as an example of a paper explicitly studying structural liveness for a specific subclass). In particular, we use established results regarding liveness for conservative nets, for which we refer to [MW14].

Organization of the paper. Section 2 introduces basic notions and notation, defines linear systems, and states the main results; it also establishes the equivalence of conservativeness and structural boundedness for reversible nets. Section 3 establishes the aforementioned 2-exp upper bound, drawing on results proven in Sections 4 and 5. Finally, Section 6 addresses the EXPSPACE lower bound.

We note that throughout this paper, we adopt a convention where lemmas are considered more significant than propositions.

2. Basic Definitions, and Results

By $\mathbb{N}$, $\mathbb{N}_+$, $\mathbb{Z}$, and $\mathbb{R}$ we denote the sets of nonnegative integers, positive integers, integers, and reals, respectively. For $i, j \in \mathbb{Z}$ we put $[i, j] = \{i, i+1, \ldots, j\}$. The unary operation $|.|$ denotes the absolute value for numbers, the cardinality for sets, and the length for sequences. For a vector $x \in \mathbb{Z}^d$ ($d \in \mathbb{N}$), we denote by $x(i)$ the value of its component $i \in [1, d]$, hence $x = (x(1), x(2), \ldots, x(d))$. We use the *component-wise (partial) order* $\leq$ on $\mathbb{Z}^d$.

Vectors, restrictions. It will be clear from context when a vector is understood as a column vector; for example, if B is an $m \times n$ matrix, then in the equation $Bx = b$ the vectors x and b are viewed as column vectors of dimensions n and m, respectively. By $\mathbf{0}$ we denote the zero vector, whose dimension is clear from context. Sometimes we consider vectors as elements of $\mathbb{Z}^J$, where J is a finite subset of $\mathbb{N}$; for example, $\mathbb{N}^d$ is identified with $\mathbb{N}^{[1,d]}$. For $x \in \mathbb{Z}^d$ and $J \subseteq [1, d]$, we denote by $x_{|J}$ the restriction of x to J, that is, the vector in $\mathbb{Z}^J$ satisfying $x_{|J}(i) = x(i)$ for each $i \in J$. For $X \subseteq \mathbb{Z}^d$, we put $X_{|J} = \{x_{|J} \mid x \in X\}$.

Multiplication, dot product, rank. We use "$\cdot$" for standard multiplication (including multiplication of a vector or a matrix by a scalar), but for $x, y \in \mathbb{Z}^d$ we use $(x \cdot y)$ to denote the dot product $\sum_{i=1}^{d} x(i) \cdot y(i)$. The *rank* of an $m \times n$ matrix B is denoted by $\mathrm{rank}(B)$; in particular, $\mathrm{rank}(B) \leq \min\{m, n\}$.

Norms for vectors, sets, and matrices. We use the following *norms* for vectors $x \in \mathbb{Z}^J$ and finite sets $X \subseteq \mathbb{Z}^J$ (where $\mathbb{Z}^J$ may in particular be $\mathbb{Z}^d$):

$$||x|| = \max_{i \in J} |x(i)|, \quad \|x\|_1 = \sum_{i \in J} |x(i)|, \quad ||X|| = \max_{x \in X} ||x||, \quad \|X\|_1 = \max_{x \in X} \|x\|_1;$$

we stipulate that $\max \emptyset = 0$. Hence,

$$0 \leq ||x|| \leq \|x\|_1 \leq |J| \cdot ||x||, \text{ and } 0 \leq ||X|| \leq \|X\|_1 \leq |J| \cdot ||X||.$$

We also define $||B||$ and $\|B\|_1$ *for matrices over* $\mathbb{Z}$, viewing a matrix as the set of its rows. For an $m \times n$ matrix B, we have $\|B\| = \max_{i,j} |B_{ij}|$ and $\|B\|_1 = \max_{i \in [1,m]} \sum_{j=1}^{n} |B_{ij}|$.

Submononoid $\boldsymbol{X^*}$. For $X \subseteq \mathbb{Z}^d$, we denote by X^* the *submonoid of* $(\mathbb{Z}^d, +)$ *generated by* X, that is, the set of all finite sums of elements of X.

Petri nets. A common definition presents a Petri net as a tuple (P, T, W) where P and T are finite disjoint sets of *places* and *transitions*, respectively, and $W \colon (P \times T) \cup (T \times P) \to \mathbb{N}$ is viewed as a set of *weighted arcs*. Figure 1 depicts places as circles, transitions as boxes, and nonzero-weight arcs, while leaving weights equal to 1 implicit; an arc with arrows at both ends denotes two arcs, one in each direction. Nevertheless, we adopt an equivalent definition that we find notationally more convenient in our context.

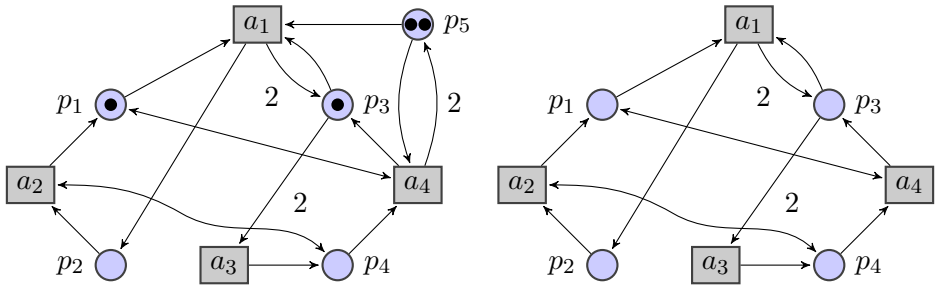


FIGURE 1. A 5-dim net A_1 (left), and a 4-dim net A_2 (right), a restriction of A_1.

Petri nets as sets of actions. A *Petri net A of dimension $d \in \mathbb{N}$*, a *d-dim net A* for short, is a finite set of pairs $a = (a_-, a_+) \in \mathbb{N}^d \times \mathbb{N}^d$ which are referred to as *actions* (a synonym for *transitions*). Each action $a = (a_-, a_+)$ has an associated *displacement*, defined as $\Delta(a) = (a_+ - a_-) \in \mathbb{Z}^d$. We set

$$||A|| = \max_{a \in A} ||a||, \text{ where } ||a|| = ||\{a_+, a_-\}||.$$

We note the close relation to *vector addition systems* (VASs), where a Petri net action $a = (a_-, a_+)$ is replaced by the plain vector $\Delta(a)$.

Markings, and relations $\xrightarrow{\boldsymbol{a}}$, $\dashrightarrow^{\boldsymbol{a}}$. A *marking* of A is a vector $x \in \mathbb{N}^d$ assigning the value $x(i)$, that is, the number of *tokens*, to each *component* (a synonym for *place*) $i \in [1, d]$. An *action a* is *enabled at a marking x* if $x \geq a_-$. The "one-step" *relation* $\xrightarrow{a}$ is defined on $\mathbb{N}^d$ by setting $x \xrightarrow{a} y$ if $x \geq a_-$ and $y = x + \Delta(a) = x - a_- + a_+$. The *virtual* "one-step" *relation* $\dashrightarrow^{a}$ is defined on $\mathbb{Z}^d$ by setting $x \dashrightarrow^{a} y$ if $y - x = \Delta(a)$. Hence, $x \xrightarrow{a} y$ implies $x \dashrightarrow^{a} y$, whereas the converse, even for $x, y \in \mathbb{N}^d$, does not necessarily hold.

Conservative nets. A d-dim *net A* is *conservative* if there exists $w \in (\mathbb{N}_+)^d$ such that $(w \cdot \Delta(a)) = 0$ for all $a \in A$; in this case $x \xrightarrow{a} y$ implies $(w \cdot x) = (w \cdot y)$. If, moreover, $w(i) = 1$ for all $i \in [1, d]$, then A is *1-conservative*. Note that in a 1-conservative net, performing actions preserves the total number of tokens, whereas in a conservative net, only a *weighted* sum of tokens is maintained.

Example 2.1. Figure 1(left) shows a 5-dim net A_1 with 4 actions. For instance, $a_1 = ((a_1)_-, (a_1)_+) = ((1,0,1,0,1), (0,1,2,0,0))$, $\Delta(a_1) = (-1,1,1,0,-1)$, $\Delta(a_2) = (1,-1,0,0,0)$, $\Delta(a_3) = (0,0,-2,1,0)$. We have $(0,2,0,1,0) \xrightarrow{a_2} (1,1,0,1,0)$, and $(0,2,0,0,0) \dashrightarrow^{a_2} (1,1,0,0,0)$, but *not* $(0,2,0,0,0) \xrightarrow{a_2} (1,1,0,0,0)$. The net A_1 is conservative, as demonstrated by the vector $w = (1,1,1,2,1)$. The net A_2 in Figure 1(right) is obtained from A_1 by removing the fifth component, that is, place p_5. Note that A_2 is not conservative; indeed, $\Delta(a_1 a_2) = (0,0,1,0)$ precludes the existence of a preserved weighted sum with positive component weights.

The relation $\xrightarrow{\boldsymbol{\sigma}}$, reachability set $\mathbf{Reach}(\boldsymbol{x})$, virtual reachability relation $\dashrightarrow^{\boldsymbol{\sigma}}$. Given a d-dim net A, we define the relation $\xrightarrow{\sigma} \subseteq \mathbb{N}^d \times \mathbb{N}^d$ for any action sequence $\sigma = a_1 a_2 \ldots a_k$ as the composition $\xrightarrow{a_1} \circ \xrightarrow{a_2} \cdots \circ \xrightarrow{a_k}$. The *displacement of σ* is given by $\Delta(\sigma) = \sum_{j=1}^{k} \Delta(a_j)$. We also refer to $x \xrightarrow{\sigma} y$ as an *execution in A*, *from* a marking x *to* a marking y.

The *reachability relation in A* is $\xrightarrow{*} \subseteq \mathbb{N}^d \times \mathbb{N}^d$, where $x \xrightarrow{*} y$ holds if $x \xrightarrow{\sigma} y$ for some action sequence σ. The *reachability set of* a marking x is defined as $\mathrm{Reach}(x) = \{y \mid x \xrightarrow{*} y\}$. We also define the relation $\dashrightarrow^{\sigma} \subseteq \mathbb{Z}^d \times \mathbb{Z}^d$ by setting $x \dashrightarrow^{\sigma} y$ if $y - x = \Delta(\sigma)$, and we refer to $x \dashrightarrow^{\sigma} y$ as a *virtual execution in*

A. The *virtual reachability relation in* A is $\overset{*}{\dashrightarrow} \subseteq \mathbb{Z}^d \times \mathbb{Z}^d$, where $x \overset{*}{\dashrightarrow} y$ holds if $x \overset{\sigma}{\dashrightarrow} y$ for some action sequence σ. We set

$$A_\delta = \{\Delta(a) \mid a \in A\},$$

and note that $x \overset{*}{\dashrightarrow} y$ if and only if $(y - x)$ belongs to the submonoid $(A_\delta)^*$ of $(\mathbb{Z}^d, +)$. We further write A_δ^* instead of $(A_\delta)^*$.

Reversible nets. A net A is *structurally reversible*, referred to as *reversible* in this paper, if the monoid A_δ^* is a subgroup of $(\mathbb{Z}^d, +)$, that is, if for every $a \in A$ we have $-\Delta(a) \in A_\delta^*$. Note that A is reversible if and only if virtual reachability is symmetric ($x \overset{*}{\dashrightarrow} y$ iff $y \overset{*}{\dashrightarrow} x$).

Restricted nets and their executions. For a d-dim net A and $I \subseteq [1, d]$,

$$\text{the } |I|\text{-dim net } A_{|I}$$

is the *restriction of* A *to* the *components* (with indices) in I. Each action $a = (a_-, a_+)$ of A gives rise to an action $a_{|I} = (a_{-|I}, a_{+|I})$ of $A_{|I}$. We refer to executions $x \xrightarrow{\sigma_{|I}} y$ in $A_{|I}$, implicitly assuming that $x, y \in \mathbb{N}^I$, and adopt the convention that

$$x_{|I} \xrightarrow{\sigma} y_{|I} \text{ stands for } x_{|I} \xrightarrow{\sigma_{|I}} y_{|I}.$$

Analogous conventions apply to virtual executions $x \overset{\sigma_{|I}}{\dashrightarrow} y$, $x_{|I} \overset{\sigma}{\dashrightarrow} y_{|I}$.

Remark 2.2. Our definition does not exclude the nets of dimension 0. Though these trivial nets are of no interest, they may arise in our context as $A_{|I}$ when $I = \emptyset$. Formally, the empty tuple () is the unique marking of the net $A_{|\emptyset}$, and we have $() \xrightarrow{a} ()$ for all actions $a \in A$.

Proposition 2.3 (**Restrictions preserve reversibility**).
If a d-dim net A is reversible and $I \subseteq [1, d]$, then $A_{|I}$ is reversible.

Note that this implication does not hold for conservativeness, as demonstrated by Example 2.1, where A_1 is conservative but its restriction $A_2 = (A_1)_{|[1,4]}$ is not.

Liveness, structural liveness, dead markings. A *marking* $x \in \mathbb{N}^d$ of a d-dim net A is *live* if for every action $a \in A$ and every $x' \in \mathrm{Reach}(x)$, there exists $y \in \mathrm{Reach}(x')$ such that $y \geq a_-$ (that is, a is enabled at y, allowing the step $y \xrightarrow{a} y + \Delta(a)$). A d-dim *net* A is *structurally live* if there exists a marking $x \in \mathbb{N}^d$ of A that is live. Note the following standard fact:

Proposition 2.4 (**Liveness is closed under reachability**).
If a marking x of a net A is live, then all markings $y \in \mathrm{Reach}(x)$ are also live.

An *action* $a \in A$ is *dead at* x if no marking in $\mathrm{Reach}(x)$ enables a; that is, for each $y \in \mathrm{Reach}(x)$ we have $y \not\geq a_-$. For technical convenience, we say that a *marking* x is *dead* if some *action* $a \in A$ is *dead at* x. (While in other contexts a marking is typically considered dead only if *all* actions are dead, we adopt this weaker notion throughout the paper.) With this terminology, we observe the following simple fact:

Proposition 2.5 (**Nonliveness as reachability of dead markings**).
A marking x of a net A is nonlive if and only if $\mathrm{Reach}(x)$ *contains a dead marking.*

Example 2.6. The net A_1 in Figure 1 is structurally live. Indeed, $x_0 = (1, 0, 1, 0, 2)$ is a live marking, which is easily verifiable since there is only one (infinite) execution from x_0: $(1, 0, 1, 0, 2) \xrightarrow{a_1} (0, 1, 2, 0, 1) \xrightarrow{a_3} (0, 1, 0, 1, 1) \xrightarrow{a_2} (1, 0, 0, 1, 1) \xrightarrow{a_4} (1, 0, 1, 0, 2) \xrightarrow{a_1} \cdots$.

By adding a token in p_4, we obtain a nonlive marking $x'_0 = (1,0,1,1,2)$. Specifically, $x'_0 \xrightarrow{a_1 a_2 a_1} y = (0,1,3,1,0)$, where y is dead because actions a_1 and a_4 are dead at y (due to $y(5) = 0$). We could also verify that a necessary condition for x to be live is that $x(i) < 3$ for all $i \in [1,4]$; however, there is no such bound on $x(5)$. (If $x(i) \geq 3$ for some $i \in [1,4]$, then x can reach a dead marking; on the other hand, there exist live markings x with arbitrarily large $x(5)$.)

Remark 2.7. Our example also demonstrates that liveness is not monotonic, even in conservative nets; that is, $x \leq x'$ and x being live does not necessarily imply that x' is live. In general, the set of live markings of a 1-conservative net is not even semilinear (that is, not expressible in Presburger arithmetic); the example from [JP19] can be adapted to 1-conservative nets.

Linear Systems $\boldsymbol{S}$, and their size-parameters $||\boldsymbol{S}||$, $\mathbf{lcm}(\boldsymbol{S})$. We now define systems that are, in essence, quantifier-free formulas of Presburger arithmetic with divisibility constraints. Such a formula is d-dimensional if it contains d integer variables.

Let $\mathbf{x}$ be a variable vector ranging over $\mathbb{Z}^d$, a *d-dim variable* for short, viewed as a tuple $\mathbf{x} = (\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_d)$ of d integer variables. A constraint of the form $(\alpha \cdot \mathbf{x}) \sim c$ where $\alpha \in \mathbb{Z}^d$, $c \in \mathbb{Z}$, and $\sim \in \{=, \geq\}$ is an *equality constraint* if $\sim$ is $=$, and an *inequality constraint* if $\sim$ is $\geq$. It is a *homogeneous constraint* if $c = 0$. For $m \in \mathbb{N}_+$, a constraint $(\alpha \cdot \mathbf{x}) \equiv c \,(\mathrm{mod}\, m)$, where $\alpha \in [0, m-1]^d$ and $c \in [0, m-1]$, is a *divisibility constraint*.

A *d-dim linear system* S is a propositional formula whose atomic propositions are equality, inequality, and divisibility constraints for a fixed d-dim variable. The *set of solutions* $[\![S]\!]$ consists of the vectors $x \in \mathbb{Z}^d$ satisfying S; if $[\![S]\!] \neq \emptyset$, then S is *satisfiable*.

By the *norm* $||S||$ we mean the least $s \in \mathbb{N}$ such that $\max\{||\alpha||, |c|\} \leq s$ for all equality and inequality constraints $(\alpha \cdot \mathbf{x}) \sim c$ in S. Moreover, $\mathrm{lcm}(S)$ is the *least common multiple* of all m occurring in $(\mathrm{mod}\, m)$ in divisibility constraints in S, with $\mathrm{lcm}(S) = 1$ if no such constraints exist.

Small solution theorems. When dealing with solutions of linear systems, *small-solution theorems* provide bounds on the size of minimal solutions. Given a set $M \subseteq \mathbb{N}^d$, by $\min_{\leq}(M)$ we denote the set of minimal vectors in M with respect to the component-wise order $\leq$. Since $\leq$ is a well-partial-order on $\mathbb{N}^d$, the set $\min_{\leq}(M)$ is always finite. In [Pot91], Pottier provided several bounds for minimal nonnegative solutions of a conjunction of homogeneous equality constraints. We now recall a bound that is particularly useful for our purposes.

Lemma 2.8 [Pot91] (**Homogeneous equality constraints**).
Let $M = \{x \in \mathbb{N}^n \mid Bx = \mathbf{0}\}$ where B is an $m \times n$ matrix over $\mathbb{Z}$. Then $X = \min_{\leq}(M \setminus \{\mathbf{0}\})$ is a finite set such that $M = X^*$. Moreover the following bound holds, where $r = \mathrm{rank}(B)$:

$$\|X\|_1 \leq (1 + \|B\|_1)^r.$$

Note that the rank bound r in Lemma 2.8 satisfies $r \leq \min\{m, n\}$. There is also a bound on the solutions of a conjunction of inequality constraints, $Bx \geq b$, in [Pot91], but it involves an exponent of m. This is not convenient for us when m, the number of constraints, is much larger than $n = d$, the dimension of our problem. Therefore in Section 4, we provide a proof of the following theorem:

Theorem 2.9 (**Bounds on solutions of linear systems**).
Every satisfiable d-dim linear system S has a solution $x \in \mathbb{Z}^d$ such that

$$||x|| \leq \mathrm{lcm}(S) \cdot \left(1 + (d+1)\,! \cdot (d \cdot ||S|| + 1)^d\right).$$

Remark 2.10. Later, in the proof of Theorem 2.13, we employ a d'-dim linear system S with $d' = d \cdot 2^d$. It is crucial for our purposes that Theorem 2.9 then yields a solution that is at most doubly exponential (2-exp) in d. In contrast, using [Kla08, Theorem 3.12], we could only derive that for S (even without divisibility constraints), the size of the minimal automaton encoding $[\![S]\!]$ in binary is bounded by $(2 + 2 \cdot ||S||)^{|S|}$, where $|S|$ is the number of

constraints in S. In our case, $|S| \geq 2^d$, and thus the bounds on the minimal solutions of S derived from the shortest accepting paths in such an automaton would be at least $2^{(2+2\cdot||S||)^{|S|}}$, which is triple-exponential (3-exp) in d.

Linear Systems for Subgroups of $\mathbb{Z}^{\boldsymbol{d}}$. In the sequel, by a *group* we mean a *subgroup* of $(\mathbb{Z}^d, +)$, which is also called a *lattice* in this context. The group *spanned* by a set $X \subseteq \mathbb{Z}^d$ is the monoid $(X \cup -X)^*$, that is, the set of finite integer linear combinations of elements of X. In Section 5, we prove the following theorem, which provides a way to encode any such group by a linear system, with a bound on its size.

Theorem 2.11 (**Groups expressed by linear systems of bounded size**)**.**
Let L be the group spanned by a finite set $X \subseteq \mathbb{Z}^d$. There exists a d-dim linear system S such that

$$[\![S]\!] = L \text{ and } \max\{||S||, \mathrm{lcm}(S)\} \leq d! \cdot \|X\|^d.$$

We thus obtain a corollary characterizing the virtual reachability of reversible nets, recalling that $A_\delta = \{\Delta(a) \mid a \in A\}$:

Corollary 2.12 (**Reversible virtual reachability expressed by a linear system**)**.**
For every reversible d-dim net A, there exists a $2d$-dim linear system S_A such that

$$[\![S_A]\!] = \{(x, y) \in \mathbb{Z}^d \times \mathbb{Z}^d \mid x \xrightarrow{*} y\}, \text{ and } \max\{||S_A||, \mathrm{lcm}(S_A)\} \leq d! \cdot ||A_\delta||^d.$$

Proof. We recall that $x \xrightarrow{*} y$ iff $y - x \in A_\delta^*$, and that A_δ^* is a subgroup of $(\mathbb{Z}^d, +)$ since A is reversible. Let S be the linear system guaranteed by Theorem 2.11 for the group $L = A_\delta^*$ (spanned by the set A_δ) with a d-dim variable $\mathbf{x}' = (\mathbf{x}'_1, \mathbf{x}'_2, \ldots, \mathbf{x}'_d)$.

We construct S_A with a $2d$-dim variable $(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_d, \mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_d)$ by replacing every occurrence of $\mathbf{x}'_i$ in S with the expression $(\mathbf{y}_i - \mathbf{x}_i)$. It follows that

$$[\![S_A]\!] = \{(x, y) \in \mathbb{Z}^d \times \mathbb{Z}^d \mid y - x \in [\![S]\!]\},$$

which is precisely the set of pairs (x, y) such that $x \xrightarrow{*} y$. Since the transformation only involves substituting a difference of variables for a single variable, the norms and the lcm remain unchanged: $||S_A|| = ||S||$, $\mathrm{lcm}(S_A) = \mathrm{lcm}(S)$. The claimed bound $d! \cdot ||A_\delta||^d$ then follows directly from Theorem 2.11. □

We will employ Theorem 2.9 and Corollary 2.12 to prove our main result, Theorem 2.13. Before stating the result, we introduce some further relevant notions.

Live bottom SCCs in reachability graphs. Given a d-dim net A, we denote by $(\mathbb{N}^d, \to)$ the *reachability graph* of A, where $x \to y$ if $x \xrightarrow{a} y$ for some $a \in A$. A nonempty set $X \subseteq \mathbb{N}^d$ such that $\mathrm{Reach}(x) = X$ for all $x \in X$ corresponds to a *bottom strongly connected component* in the graph; we simply call X a *bottom SCC* of A in this case. A bottom SCC X is *live* if some $x \in X$ is live (which implies that all $x \in X$ are live, by Proposition 2.4).

1-exp and 2-exp functions. We call a function $f\colon \mathbb{N}^2 \to \mathbb{N}$ a *1-exp* function if

$$f(m, i) \leq (2 + m)^{\mathrm{poly}(i)},$$

meaning that there exists a polynomial $p\colon \mathbb{N} \to \mathbb{N}$ such that $f(m, i) \leq (2 + m)^{p(i)}$ for all $m, i \in \mathbb{N}$. It is a *2-exp* function if

$$f(m, i) \leq (2 + m)^{2^{\mathrm{poly}(i)}}.$$

Throughout the paper, we specifically use values of the form $f(||A||, d)$ or $f(||A||, |I|)$ for $I \subseteq [1, d]$, where A is a d-dim net.

Theorem 2.13 (**Bounds on minimal live markings in nets with live bottom SCCs**)**.** *There exists a 2-exp function f such that for every d-dim net A possessing a live bottom SCC, there exists a live marking $x \in \mathbb{N}^d$ satisfying $||x|| \leq f(||A||, d)$.*

Live bottom SCCs, reversibility, conservativeness, structural boundedness. We say that a *marking* $x \in \mathbb{N}^d$ of a d-dim net A is *bounded* if the set of reachable markings $\mathrm{Reach}(x) = \{y \mid x \xrightarrow{*} y\}$ is finite; that is, there exists $B_x \in \mathbb{N}$ such that $||y|| \leq B_x$ for all $y \in \mathrm{Reach}(x)$. A *net* A is *structurally bounded* if every marking x of A is bounded.

The following lemma shows that structurally live conservative nets coincide with structurally live structurally bounded nets, and that Theorem 2.13 applies to these nets.

Lemma 2.14 (**Conservativeness vs. structural boundedness for live nets**)**.**

(1) Every conservative net is structurally bounded.
(2) Every structurally bounded net that is structurally live has a live bottom SCC.
(3) Every net possessing a live bottom SCC is reversible.
(4) Every structurally bounded net that is reversible is conservative.

Proof. (1) If a d-dim net A is conservative, there exists $w \in (\mathbb{N}_+)^d$ such that $(w \cdot \Delta(a)) = 0$ for all $a \in A$. It follows that $x \xrightarrow{*} y$ implies $(w \cdot x) = (w \cdot y)$. Since all components of w are positive, the set $C_x = \{y \in \mathbb{N}^d \mid (w \cdot y) = (w \cdot x)\}$ is finite for any $x \in \mathbb{N}^d$. Since $\mathrm{Reach}(x) \subseteq C_x$, the net A is structurally bounded.

(2) Let x be a live marking of a structurally bounded net A. Since x is bounded, $\mathrm{Reach}(x)$ is finite, which implies that there must exist a (finite) bottom SCC $X \subseteq \mathrm{Reach}(x)$ in the reachability graph of A. Since x is live, every $y \in \mathrm{Reach}(x)$ is also live (by Proposition 2.4); consequently, X is a live bottom SCC of A.

(3) Let X be a (finite or infinite) live bottom SCC of a net A. By the definition of liveness, for each action $a \in A$, there exists some $y \in X$ at which a is enabled, that is, $y \xrightarrow{a} y + \Delta(a)$. Since X is a bottom SCC, we must have $y + \Delta(a) \xrightarrow{*} y$. This implies that $y - (y + \Delta(a)) = -\Delta(a) \in A^*_\delta$. Therefore, $-\Delta(a) \in A^*_\delta$ for every $a \in A$, which shows that A is reversible.

(4) Let $A = \{a_1, a_2, \ldots, a_k\}$ be a d-dim net that is structurally bounded and reversible. For each $i \in [1, d]$, let $e_i \in \mathbb{N}^d$ be the ith unit vector (satisfying $e_i(i) = 1$ and $e_i(j) = 0$ for all $j \neq i$). We observe that there exists no $v \in \mathbb{N}^k$ such that

$$\sum_{j=1}^{k} v(j) \cdot \Delta(a_j) \;\geq\; e_i. \tag{1}$$

Indeed, if such v existed, then for an action sequence σ with Parikh vector v (*e.g.*, $\sigma = (a_1)^{v(1)} \ldots (a_k)^{v(k)}$) and a sufficiently large marking x, we would have $x \xrightarrow{\sigma} x + \Delta(\sigma)$ with $\Delta(\sigma) \geq e_i$. This would imply $x \xrightarrow{\sigma^n} x + n \cdot \Delta(\sigma) \geq x + n \cdot e_i$ for all $n \in \mathbb{N}$, contradicting structural boundedness.

Since A is reversible, every $\Delta(a_j)$ can be "undone" by some action sequence; that is, $-\Delta(a_j) = \sum_{\ell=1}^{k} u(\ell) \cdot \Delta(a_\ell)$ for some $u \in \mathbb{N}^k$. Hence every integer combination of the vectors $\Delta(a_j)$ equals some nonnegative one, so (1) has no solution $v \in \mathbb{Z}^k$ either. Finally, the $\Delta(a_j)$ are integer vectors, so if (1) has a real solution then it has a rational one, and hence an integer one after multiplying by a common denominator. Therefore (1) has no solution $v \in \mathbb{R}^k$ either.

Now we recall the following variant of Farkas' lemma (see, *e.g.*, Proposition 6.4.3(iii) in [MG07]):

> Let M be a real matrix with m rows and n columns, and let $b \in \mathbb{R}^m$ be a vector. The system $Mx \leq b$ has a solution $x \in \mathbb{R}^n$ if and only if every nonnegative $y \in \mathbb{R}^m$ with $y^{\mathsf{T}} M = \mathbf{0}^{\mathsf{T}}$ also satisfies $y^{\mathsf{T}} b \geq 0$.

Fix $i \in [1,d]$, let M be the $d \times k$ matrix whose jth column is $-\Delta(a_j)$, and let $b = -e_i$. Then $Mv \leq b$ is precisely (1), which has no solution $v \in \mathbb{R}^k$. By the lemma, there exists a nonnegative $y \in \mathbb{R}^d$ with $y^\mathsf{T} M = \mathbf{0}^\mathsf{T}$ and $y^\mathsf{T} b < 0$. The jth entry of $y^\mathsf{T} M$ is $(y \cdot -\Delta(a_j))$, so the former condition amounts to $(y \cdot \Delta(a_j)) = 0$ for all $j \in [1,k]$; and $y^\mathsf{T} b = (y \cdot -e_i) = -y(i)$, so the latter amounts to $y(i) > 0$. By the rationality argument above, y may be chosen to be a rational vector, and hence an integer vector after multiplying by a common denominator; both conditions are preserved. Writing w_i for the resulting vector, we obtain for each $i \in [1,d]$ some $w_i \in \mathbb{N}^d$ with $w_i(i) > 0$ and $(w_i \cdot \Delta(a_j)) = 0$ for all $j \in [1,k]$.

Setting $w = \sum_{i=1}^{d} w_i$ yields a vector $w \in (\mathbb{N}_+)^d$ satisfying $(w \cdot \Delta(a_j)) = 0$ for all $j \in [1,k]$. Consequently, the net A is conservative. □

The EXPSPACE-hardness result from [JP19] will be slightly strengthened by Theorem 6.1 in Section 6; Theorem 2.13 thus allows us to obtain a matching upper bound.

Corollary 2.15 (EXPSPACE-completeness of structural liveness).

(1) Structural liveness for conservative nets is EXPSPACE-complete.
(2) Structural liveness for structurally bounded nets is EXPSPACE-complete.

Proof. Since EXPSPACE=NEXPSPACE, it suffices to provide a nondeterministic algorithm working in exponential space that verifies positive instances of the considered problems. By Lemma 2.14, the positive instances of both problems coincide; in particular, every structurally live structurally bounded net is conservative.

Given a structurally live conservative d-dim net A, an algorithm can guess a live marking $x \in \mathbb{N}^d$ as guaranteed by Theorem 2.13. Since $||x|| \leq f(||A||, d)$ for the corresponding 2-exp function f satisfying $f(m,i) \leq (2+m)^{2^{\mathrm{poly}(i)}}$, the binary representation of x has size at most $O(2^{\mathrm{poly}(d)} \cdot \log(2 + ||A||))$, which fits in exponential space. Finally, verifying the liveness of x for A can be performed in polynomial space with respect to the binary encoding of A and x (see, *e.g.*, [MW14, Lemma 5]), which remains within exponential space relative to the original input size. □

3. Upper Bound (Proof of Theorem 2.13)

In Section 3.1, a proof of Theorem 2.13 is presented; the proofs of some auxiliary results are deferred to Sections 3.2 and 3.3. We begin with several observations and conventions.

1-exp and 2-exp functions. We have defined 1-exp and 2-exp functions as functions $f \colon \mathbb{N}^2 \to \mathbb{N}$ satisfying

$$f(m,i) \leq (2+m)^{\mathrm{poly}(i)} \text{ and } f(m,i) \leq (2+m)^{2^{\mathrm{poly}(i)}},$$

respectively. For technical convenience, we implicitly assume that each such function f that we consider is *non-decreasing* in both arguments; that is, $m \leq m'$ and $i \leq i'$ imply $f(m,i) \leq f(m',i')$. Furthermore, we assume that f is *extensive*, meaning that $m \leq f(m,i)$.

It is straightforward to verify that the class of 1-exp functions, as well as the class of 2-exp functions, is closed under standard operations such as sum, product, and composition, where the latter is defined as

$$(f \circ g)(m,i) = f(g(m,i),i).$$

For instance, if $f(m,i) \leq (2+m)^{p_1(i)}$ and $g(m,i) \leq (2+m)^{p_2(i)}$ with $p_1(i) \geq 1$ and $p_2(i) \geq 1$, then

$$(f \circ g)(m,i) \leq \left(2 + (2+m)^{p_2(i)}\right)^{p_1(i)} \leq (2+m)^{2 \cdot p_2(i) \cdot p_1(i)}.$$

The functions p_1, p_2 can be, in particular, polynomials or exponential functions.

We define the j-th iterate of f recursively as follows:

$$f^{(0)}(m,i) = m, \text{ and } f^{(j+1)}(m,i) = f(f^{(j)}(m,i),i).$$

Note that if $f(m,i)$ is a 1-exp function or a 2-exp function, then $f'(m,i) = f^{(i)}(m,i)$ is a 2-exp function. Indeed, if $f(m,i) \leq (2+m)^{p(i)}$, where p is a polynomial or exponential function, then

$$f'(m,i) = f^{(i)}(m,i) \leq (2+m)^{2^{i-1}\cdot(p(i))^i} = (2+m)^{2^{(i-1)+i\cdot\log p(i)}} \leq (2+m)^{2^{\mathrm{poly}(i)}}.$$

We use these closure properties implicitly when deriving the existence of specific 1-exp or 2-exp functions throughout this section.

1-exp and 2-exp constants. For convenience, when considering a d-dim net A, we also refer to *1-exp* and *2-exp constants*. We say that C is a 1-exp (resp. 2-exp) constant related to A if $C = f(||A||, d)$ for some 1-exp (resp. 2-exp) function that is independent of A. For instance, we will use a 2-exp constant C^A_{dead} associated with a 2-exp function f_{dead}; the superscript A clarifies that $C^A_{\mathrm{dead}} = f_{\mathrm{dead}}(||A||, d)$, where d is the dimension of the net A.

3.1. **Proof of Theorem 2.13.**

We aim to show the existence of a 2-exp function f such that for every d-dim net A possessing a live bottom SCC $X \subseteq \mathbb{N}^d$ (where every $x \in X$ is a live marking and $\mathrm{Reach}(x) = X$ for all $x \in X$), there exists a live marking $x \in \mathbb{N}^d$ satisfying $||x|| \leq f(||A||, d)$. Recall that such a net A is reversible by Lemma 2.14.

In this section, we provide the overall proof, deferring certain technical parts to the subsequent sections.

Remark 3.1. Some technical issues are addressed by referring to results from [Ler19]. Specifically, these concern d-dimensional *extractors* (used to distinguish between small and large components of markings, or of sets of markings), a 1-exp *virtual reachability* function f_{vr} for reversible Petri nets with states (PNSs), and a 2-exp *mutual-reachability* function f_{mr} for general Petri nets.

Dead markings, 2-exp constant $\boldsymbol{C^A_{\mathbf{dead}}}$. Recall that a marking $x \in \mathbb{N}^d$ of a d-dim net A is live if and only if there is no dead marking in $\mathrm{Reach}(x)$ (Proposition 2.5). By our convention, a marking y is dead if at least one action a is dead at y, that is, $y' \not\geq a_-$ for all $y' \in \mathrm{Reach}(y)$. Our goal is to bound the minimal size of solutions $x \in \mathbb{N}^d$ to the constraint

(2) "for each dead marking y, we have $x \not\xrightarrow{*} y$".

In Section 3.2 we recall, by Lemma 3.36 using Rackoff's result in [Rac78], that

the set of dead markings of a d-dim net A

can be presented as the set of solutions $y \in \mathbb{N}^d$ of a disjunction

$$\bigvee_{I \subseteq [1,d]} \bigvee_{j=1}^{k_I} (\mathbf{y}_{|I} = \alpha_{(I,j)}) \tag{3}$$

where each (restricted) marking $\alpha_{(I,j)} \in \mathbb{N}^I$ is *dead in the restricted net* $A_{|I}$ and satisfies

$$||\alpha_{(I,j)}|| < C^A_{\mathrm{dead}},$$

for some 2-exp constant C^A_{dead}. We allow $k_I = 0$ for some $I \subseteq [1,d]$; the corresponding disjunction $\bigvee_{j=1}^{0}$ is empty, and thus false. The disjunction (3) can therefore be restricted to $\bigvee_{I\in\mathcal{D}} \cdots$, where $\mathcal{D} = \{I \subseteq [1,d] \mid k_I > 0\}$. Note that $\emptyset \notin \mathcal{D}$: in the restricted net $A_{|\emptyset}$ no action is ever dead (recall Remark 2.2), hence no dead $\alpha_{(\emptyset,j)}$ exists.

Remark 3.2. By our convention, the expression "some 2-exp constant C^A_{dead}" means that there exists a non-decreasing function $f_{\mathrm{dead}}(m,i) \leq (2+m)^{2^{\mathrm{poly}(i)}}$ such that $C^A_{\mathrm{dead}} = f_{\mathrm{dead}}(||A||, d)$ for any d-dim net A. A more detailed analysis would involve stating that $||\alpha_{(I,j)}|| < f_{\mathrm{dead}}(||A||, |I|)$, where $f_{\mathrm{dead}}(||A||, 1) = ||A||$, and discussing the tight relation to the coverability problem (for markings a_-). However, for our purposes, it suffices to simplify the analysis by referring only to C^A_{dead}, even when a restricted net $A_{|I}$ is considered.

Technically, we do not rely on the fact that the set of dead markings of A is downward closed (that is, if y is dead and $y' \leq y$, then y' is also dead). We only use the property that, given a marking y of a net A, if $y_{|I}$ is dead in the restricted net $A_{|I}$, then y is dead in A (Proposition 3.35).

(Virtual) reachability, 1-exp $\boldsymbol{f_{\mathbf{vr}}}$ and $\boldsymbol{C^A_{\mathbf{vr}}}$, 2-exp $\boldsymbol{C^A_{\mathbf{suf}} = \max\{C^A_{\mathbf{dead}}, C^A_{\mathbf{vr}}\}}$. Our proof of Theorem 2.13 is based on replacing the non-reachability relation $\overset{*}{\not\to}$ in constraint (2) with the virtual reachability relation $\overset{*}{\dashrightarrow}$, in a way that will be clarified later. Markings in which all components are sufficiently large will play a crucial role.

Definition 3.3 ((Sub)markings with all components above a level, $x \in \uparrow C$).
We say that a *vector* $x \in \mathbb{N}^d$ is *above* a *level* $C \in \mathbb{N}$, or simply $x \in \mathbb{N}^d$ is *above* $C \in \mathbb{N}$, denoted $x \in \uparrow C$, if $x(i) \geq C$ for all $i \in [1,d]$. A restricted vector $x_{|J}$ with $J \subseteq [1,d]$ is above C, denoted $x_{|J} \in \uparrow C$, if $x(i) \geq C$ for all $i \in J$.

The following lemma is a consequence of [Ler19, Lemma 5].

Lemma 3.4 (Virtual and standard reachability in reversible nets, f_{vr}, C^A_{vr}).
There exists a 1-exp function f_{vr} such that for every reversible d-dim net A we have:

(1) If $x \overset{*}{\dashrightarrow} y$, then $x \overset{\sigma}{\dashrightarrow} y$ for some σ satisfying $|\sigma| \leq f_{\mathrm{vr}}(||A||, d) \cdot ||y - x|| = C^A_{\mathrm{vr}} \cdot ||y - x||$.

(2) If $x \overset{*}{\dashrightarrow} y$ and $x, y \in \uparrow C^A_{\mathrm{vr}}$, then $x \overset{\sigma}{\to} y$ for some σ satisfying $|\sigma| \leq C^A_{\mathrm{vr}} \cdot ||y - x||$.

Intuitively, this can be seen by recalling that $x \overset{*}{\dashrightarrow} y$ means $y - x \in A^*_\delta$. We highlight the fact that, given a reversible d-dim net A, virtual reachability coincides with standard reachability above C^A_{vr}:

$$x \overset{*}{\dashrightarrow} y \iff x \overset{*}{\to} y, \text{ for all } x, y \in \mathbb{N}^d \cap \uparrow C^A_{\mathrm{vr}}.$$

For later convenience, we define a sufficiently large 2-exp constant:

$$C^A_{\mathrm{suf}} = \max\{C^A_{\mathrm{dead}}, C^A_{\mathrm{vr}}\}. \tag{4}$$

Segment decomposition of virtual executions in reversible nets, 1-exp $\boldsymbol{C^A_{\mathbf{shift}}}$.
In Section 3.3, we show that if $x \overset{*}{\dashrightarrow} y$ in a *reversible* d-dim net A, then there exists a virtual execution

$$x = x_0 \overset{*}{\dashrightarrow} x_1 \overset{*}{\dashrightarrow} x_2 \cdots \overset{*}{\dashrightarrow} x_k = y \tag{5}$$

consisting of *1-exp segments* that *stepwise approach the target* y from x. The segments $x_j \overset{*}{\dashrightarrow} x_{j+1}$ are 1-exp bounded in the sense that

$$0 < ||x_{j+1} - x_j|| \leq C^A_{\mathrm{shift}} \tag{6}$$

for some 1-exp constant C^A_{shift} (satisfying $C^A_{\text{shift}} = f_{\text{shift}}(||A||, d) \leq (2 + ||A||)^{\text{poly}(d)}$). By the term *stepwise approaching*, we mean that the sequence $x = x_0, x_1, \ldots, x_k = y$ satisfies:

- For each $i \in [1, d]$ with $x(i) \leq y(i)$, there exists $j_i \in [0, k]$ such that
$$x(i) < x_1(i) < \cdots < x_{j_i}(i) = x_{j_i+1}(i) = \cdots = x_k(i) = y(i);$$
- For each $i \in [1, d]$ with $x(i) > y(i)$, there exists $j_i \in [0, k]$ such that
$$x(i) > x_1(i) > \cdots > x_{j_i}(i) = x_{j_i+1}(i) = \cdots = x_k(i) = y(i).$$

Note that Lemma 3.4(1) implies that for each segment $x_j \xrightarrow{*} x_{j+1}$ of (5) there exists σ with $|\sigma| \leq C^A_{\text{vr}} \cdot C^A_{\text{shift}}$ such that $x_j \xrightarrow{\sigma} x_{j+1}$, and even $x_j \xrightarrow{\sigma} x_{j+1}$ if $x_j, x_{j+1} \in {\uparrow} C^A_{\text{vr}}$.

Small and large components of vectors, relative to extractors. We need to distinguish between relatively small and large components of vectors $x \in \mathbb{N}^d$ (markings of a net A). To identify small components, we employ the notion of extractors, following [Ler19].

Definition 3.5 (**Extractors $\boldsymbol{\lambda}$ of dimension $\boldsymbol{d}$**).
For $d \geq 1$, by a *d-dim extractor* λ we mean a tuple
$$(\lambda_1, \lambda_2, \ldots, \lambda_d) \in (\mathbb{N}_+)^d \text{ where } \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_d.$$
For technical convenience, we also refer to the values λ_0 and λ_{d+1}, by implicitly setting $\lambda_0 = 1$ and $\lambda_{d+1} = \lambda_d$.

The following claim is straightforward (for further details see [Ler19]).

Proposition 3.6 (**The set $\boldsymbol{I^\lambda_x \subseteq [1, d]}$ of (indices of) small components of $\boldsymbol{x \in \mathbb{N}^d}$**).
Given a d-dim extractor λ, for each $x \in \mathbb{N}^d$ there exists a unique maximal set $I^\lambda_x \subseteq [1, d]$ such that
$$\bigwedge_{i \in I^\lambda_x} \left(x(i) < \lambda_{|I^\lambda_x|} \right) \ \wedge \bigwedge_{i \in [1,d] \setminus I^\lambda_x} \left(x(i) \geq \lambda_{|I^\lambda_x|+1} \right).$$
More generally, for each subset $J \subseteq [1, d]$, there is a unique maximal set $I^{(\lambda,J)}_x \subseteq J$ such that
$$\bigwedge_{i \in I^{(\lambda,J)}_x} \left(x(i) < \lambda_{|I^{(\lambda,J)}_x|} \right) \ \wedge \bigwedge_{i \in J \setminus I^{(\lambda,J)}_x} \left(x(i) \geq \lambda_{|I^{(\lambda,J)}_x|+1} \right).$$

For instance, if $\lambda = (3, 17, 258, 1500)$ and $x = (11, 1600, 2, 300)$, $x' = (16, 5000, 15, 5000)$, then $I^\lambda_x = I^\lambda_{x'} = \{1, 3\}$. The values $x(i)$ of the components $i \in I^\lambda_x$ are viewed as *small*, while the values $x(i)$ for $i \in [1, d] \setminus I^\lambda_x$ are *large*. The gap between the values of large components and of small components of x is bigger than $\lambda_{j+1} - \lambda_j$, where $j = |I^\lambda_x|$. In our example, with $|I^\lambda_x| = 2$, the gap is bigger than $\lambda_3 - \lambda_2 = 258 - 17 = 241$.

Figure 2 (left) depicts a marking $x \in \mathbb{N}^d$ and highlights the index set $I = I^\lambda_x \subseteq [1, d]$. The marking x is also an example of a marking above the level C^A_{suf} (defined in (4)); such markings are particularly important in our later reasoning.

Note that $I^\lambda_x = [1, d]$ if and only if $x(i) < \lambda_d$ for every $i \in [1, d]$; the condition $x(i) \geq \lambda_{d+1}$ holds vacuously for all $i \in \emptyset$. If $J' = \{i \in [1, d] \mid x(i) \geq \lambda_d\}$, then $I^\lambda_x = I^{(\lambda,J)}_x$ for $J = [1, d] \setminus J'$. Specifically, if $J' = [1, d]$, then $I^\lambda_x = \emptyset$; the condition $x(i) < \lambda_0$ holds vacuously for all $i \in \emptyset$. In general, $I^\lambda_x = \emptyset$ if and only if for each $\emptyset \neq I \subseteq [1, d]$, there exists $i \in I$ such that $x(i) \geq \lambda_{|I|}$.

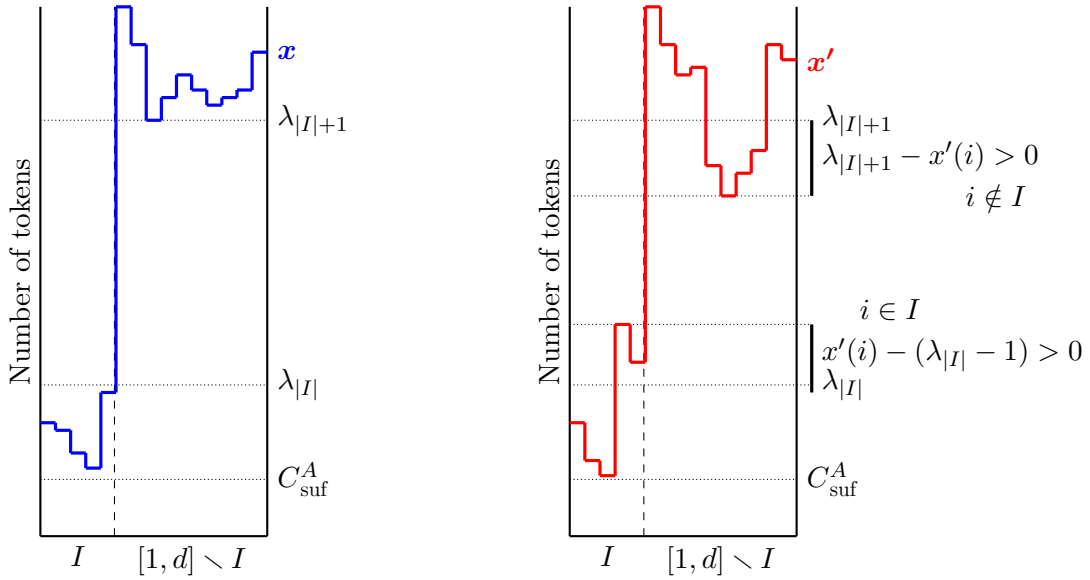


FIGURE 2. Index set $I = I^\lambda_x$ (left); depiction of (λ, I)-dist(x') (right).

Later, we apply *2-exp d-dim extractors* λ to the d-dim nets A under consideration, meaning that

$$\lambda_d = f(||A||, d) \leq (2 + ||A||)^{2^{\text{poly}(d)}}$$

for some (implicit) 2-exp function $f(m, i)$ which is independent of A. We use such extractors for which the gaps $\lambda_{j+1} - \lambda_j$ between the large and small components grow sufficiently with the index j. However, before defining specific extractors, we introduce certain linear systems with 2-exp solutions.
Linear systems underlying the proof of Theorem 2.13. In Definition 3.8 we introduce a class of linear systems on which our proof rests; Proposition 3.9 shows that these systems have 2-exp solutions.

The definition uses the notion of the *distance of a vector to a λ-case*, denoted (λ, I)-dist(x) for a vector $x \in \mathbb{N}^d$ and a λ-case I, where $\emptyset \neq I \subseteq [1, d]$. This distance is a nonnegative integer, and it is zero whenever $I^\lambda_x = I$. See Figure 2 (right) for an illustration.

Definition 3.7 (Distance (λ, I)-dist(x) of $x \in \mathbb{N}^d$ to the λ-case $\emptyset \neq I \subseteq [1, d]$).
Given a d-dim extractor λ, the *distance of* $x \in \mathbb{N}^d$ *to a λ-case* I, where $\emptyset \neq I \subseteq [1, d]$, is

$$(\lambda, I)\text{-dist}(x) = \max\left(\{0\} \cup \{x(i) - (\lambda_{|I|} - 1) \mid i \in I\} \cup \{\lambda_{|I|+1} - x(i) \mid i \in [1, d] \setminus I\}\right).$$

The value (λ, I)-dist(x) is positive if and only if there exists $i \in [1, d]$ such that $i \in I$ and $x(i) \geq \lambda_{|I|}$, or $i \in [1, d] \setminus I$ and $x(i) < \lambda_{|I|+1}$. For instance, if $\lambda = (3, 17, 258, 1500)$ and $x = (11, 1600, 2, 300)$, then $(\lambda, \{1, 3\})$-dist$(x) = 0$, and $(\lambda, \{1, 2\})$-dist$(x) = \max(\{0\} \cup \{-5, 1584\} \cup \{256, -42\}) = 1584$.

The following definition of (A, x_0, λ)-linear systems aims at transforming constraint (2). For *reversible* d-dim nets A and live markings x_0, our later choice of linear systems S in this class will guarantee that every solution of S presents a nonempty set of at most 2^d *live* markings of A that are mutually

virtually reachable; in fact, they are also mutually reachable since they all are above C^A_{vr}. Hence, the non-reachability of dead markings in constraint (2) is replaced with the positive condition of virtual reachability of a specific subset of markings. The definition uses the 1-exp constant C^A_{shift} (6) and the 2-exp constant C^A_{suf} (4) related to a d-dim net A.

Definition 3.8 (**(A, x_0, λ)-linear systems for nets, markings, and 2-exp extractors**)**.** Given a reversible d-dim net A and a 2-exp extractor $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_d)$, we define the 2-exp bound

$$B = B_{(A,\lambda)} = \max\{C^A_{\mathrm{suf}}, \lambda_d + C^A_{\mathrm{shift}}\}. \tag{7}$$

Given a marking $x_0 \in \mathbb{N}^d \cap \uparrow C^A_{\mathrm{suf}}$ (that is, $\bigwedge_{i=1}^d x_0(i) \geq C^A_{\mathrm{suf}}$), for each $\emptyset \neq I \subseteq [1,d]$ let

$$x_I^{\mathrm{close}} \in \mathbb{N}^d \cap \uparrow C^A_{\mathrm{suf}}$$

be a fixed marking in $\mathrm{Reach}(x_0)$ that is above C^A_{suf} and minimizes the distance to the λ-case I, that is, (λ, I)-dist(x_I^{close}) is minimal.

Then the following linear system S of dimension $d \cdot 2^d$, with a d-dim variable $\mathbf{x} = (\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_d)$ and $2^d - 1$ d-dim variables $\mathbf{x}_I$, one for each $\emptyset \neq I \subseteq [1,d]$, is

$$\text{an } (A, x_0, \lambda)\text{-}\textit{linear system}\text{:}$$

$$(\mathbf{x} \equiv_B x_0) \wedge \bigwedge_{\emptyset \neq I \subseteq [1,d]} \big((\mathbf{x}_I \equiv_B x_I^{\mathrm{close}}) \wedge (\mathbf{x} \xrightarrow{*} \mathbf{x}_I)\big), \tag{8}$$

where $\mathbf{y} \equiv_B x$ is a shorthand for

$$\left(\bigwedge_{i \in [1,d], x(i) \leq B} (\mathbf{y}_i = x(i))\right) \wedge \left(\bigwedge_{i \in [1,d], x(i) > B} (\mathbf{y}_i \geq B+1)\right).$$

Proposition 3.9 (**(A, x_0, λ)-linear systems have 2-exp solutions**)**.**
For every (A, x_0, λ)-linear system where A has dimension d, there exists a solution

$$x \in \mathbb{N}^{d \cdot 2^d} \textit{ such that } ||x|| \leq (2 + ||A||)^{2^{poly(d)}}.$$

Proof. Let S be an (A, x_0, λ)-linear system where A has dimension d. The system S is satisfiable, since the tuple $(x_0, (x_I^{\mathrm{close}})_{\emptyset \neq I \subseteq [1,d]})$ is one of its solutions. Due to Theorem 2.9, there exists a solution $x \in [\![S]\!]$ such that

$$||x|| \leq \mathrm{lcm}(S) \cdot \left(1 + (\bar{d}+1)\,! \cdot (\bar{d} \cdot ||S|| + 1)^{\bar{d}}\right)$$

where $\bar{d} = d \cdot 2^d$. Recalling (8) and Corollary 2.12, we deduce that $\mathrm{lcm}(S) \leq d\,! \cdot ||A||^d$, and $||S|| \leq 1 + B_{(A,\lambda)} \leq (2 + ||A||)^{2^{\mathrm{poly}(d)}}$. We thus obtain

$$||x|| \leq d\,! \cdot ||A||^d \cdot \left(1 + (d \cdot 2^d + 1)\,! \cdot (d \cdot 2^d \cdot ||S|| + 1)^{d \cdot 2^d}\right),$$

which implies that $||x|| \leq (2 + ||A||)^{2^{\mathrm{poly}(d)}}$. □

Our goal now is to show certain *lower bounds* for the increasing gaps $\lambda_{j+1} - \lambda_j$ that ensure that *every solution* of an (A, x_0, λ)-linear system S constitutes *a set of live markings* of A, under the condition that x_0 is a sufficiently large *live* marking. Under this condition, the solution $(x_0, (x_I^{\text{close}})_{\emptyset \neq I \subseteq [1,d]})$ has the required property, since all markings x_I^{close} are reachable from the live marking x_0, and are thus themselves live (by Proposition 2.4). Once we establish that *all solutions* $(\bar{x}_0, (\bar{x}_I)_{\emptyset \neq I \subseteq [1,d]})$ of S possess this property, we use the fact that the minimal solutions are guaranteed to be at most 2-exp by Proposition 3.9.

This goal is achieved via Lemma 3.14, which builds on the quasi-dead marking property of *reversible* nets established in Lemma 3.13.

Quasi-dead markings. We now show that, given a *reversible* d-dim net A, every nonlive marking $x_0 \in \uparrow \max\{C^A_{\text{suf}}, ||A|| \cdot C^A_{\text{qd}}\}$, where C^A_{qd} is a certain 2-exp constant, can reach a dead marking via a *quasi-dead* marking $x \in \uparrow C^A_{\text{suf}}$. By definition, quasi-dead markings must be above the level C^A_{suf} and reach dead markings by executions of length at most C^A_{qd}. We use Figure 3 for illustration.

Definition 3.10 (**Quasi-dead markings, 2-exp function f_{qd} and constant C^A_{qd}**).
For a 2-exp function f_{qd} and a d-dim net A, a marking $x \in \mathbb{N}^d$ is f_{qd}-*quasi-dead*, or simply *quasi-dead* when f_{qd} is clear from context, if $x \in \uparrow C^A_{\text{suf}}$ and there exists an execution $x \xrightarrow{\sigma} y$ such that y is a dead marking and $|\sigma| \leq f_{\text{qd}}(||A||, d) = C^A_{\text{qd}}$.

Lemma 3.13 uses Proposition 3.12, which is based on the following involved result [Ler19, Theorem 2]; the result applies to general nets, without assuming reversibility:

Lemma 3.11 [Ler19] (**2-exp mutual reachability function f_{mr}, 2-exp constant C^A_{mr}**). *There exists a 2-exp function f_{mr} with the following property: For every d-dim net A, if $x \xrightarrow{*} y$ and $y \xrightarrow{*} x$, then there exist action sequences σ_1, σ_2 such that $x \xrightarrow{\sigma_1} y \xrightarrow{\sigma_2} x$ and*

$$|\sigma_1 \sigma_2| \leq f_{\text{mr}}(||A||, d) \cdot ||x - y|| = C^A_{\text{mr}} \cdot ||x - y||.$$

Proposition 3.12 (**Down reachability $x \xrightarrow{*} y$ with $x \geq y$**).
Given a d-dim net A and markings $x \geq y$ such that $x \xrightarrow{*} y$, there exists a marking $x' \geq x$ satisfying the following conditions:

a) $||x' - x|| \leq C^A_{\text{mr}} \cdot ||x - y||$;
b) $x' \xrightarrow{\sigma} y$ *for some* σ *such that* $|\sigma| \leq C^A_{\text{mr}} \cdot ||x - y||$;
c) $x'(i) = x(i)$ *if* $x(i) = y(i)$ *(and* $x'(i) \geq x(i)$ *if* $x(i) > y(i)$*), for all* $i \in [1, d]$.

Proof. Given a d-dim net A, and $x, y \in \mathbb{N}^d$ such that $x \geq y$ and $x \xrightarrow{*} y$, let A' be the d-dim net obtained from A by adding actions $\mathbf{e}_i$ for those $i \in [1, d]$ for which $x(i) > y(i)$, where $\mathbf{e}_i$ increments the i-th component by 1. Formally, $(\mathbf{e}_i)_- = \mathbf{0}$ and $(\mathbf{e}_i)_+ = e_i$, where e_i is the ith unit vector satisfying $e_i(i) = 1$ and $e_i(j) = 0$ for $j \neq i$.

Hence, there exist sequences σ_1, σ_2 of actions in A' satisfying

$$x \xrightarrow{\sigma_1} y \xrightarrow{\sigma_2} x \text{ in } A'; \tag{9}$$

we can take σ_1 from A due to the assumption $x \xrightarrow{*} y$ in A, whereas σ_2 can consist solely of the increment actions $\mathbf{e}_i$. By Lemma 3.11, there exist sequences σ_1, σ_2 of actions in A' that satisfy (9) and, moreover,

$$|\sigma_1 \sigma_2| \leq C^A_{\text{mr}} \cdot ||x - y||;$$

note that the actions $\mathbf{e}_i$ might occur in σ_1 as well. We safely use $C^A_{\text{mr}} = f_{\text{mr}}(||A||, d)$ instead of $C^{A'}_{\text{mr}} = f_{\text{mr}}(||A'||, d)$, since $||A'|| = ||A||$ provided $||A|| \geq 1$; the claim is trivial if $||A|| = 0$.

Any execution $x \xrightarrow{\sigma_1} y$ in A' can be clearly reorganized into the form

$$x \xrightarrow{\sigma'} x' \xrightarrow{\sigma} y,$$

where the prefix σ' is the projection of σ_1 onto the set of increment actions $\mathbf{e}_i$, and the suffix σ is the projection of σ_1 onto the set of original actions in A.

We have thus obtained $x' \geq x$ satisfying:

a) $||x' - x|| \leq |\sigma'| \leq |\sigma_1| \leq C^A_{\mathrm{mr}} \cdot ||x - y||$;
b) $x' \xrightarrow{\sigma} y$ for some sequence σ of actions in A such that $|\sigma| \leq |\sigma_1| \leq C^A_{\mathrm{mr}} \cdot ||x - y||$;
c) $x'(i) = x(i)$ if $x(i) = y(i)$ (since the action $\mathbf{e}_i$ is not in A' for such $i \in [1, d]$). □

Lemma 3.13 (**Large nonlive markings (virtually) reach quasi-dead markings**). *There exists a 2-exp function f_{qd} such that for every reversible d-dim net A and every nonlive marking $x_0 \in \uparrow \max\{C^A_{\mathrm{suf}}, ||A|| \cdot C^A_{\mathrm{qd}}\}$, where $C^A_{\mathrm{qd}} = f_{\mathrm{qd}}(||A||, d)$, there exists an f_{qd}-quasi-dead marking $x \in \mathrm{Reach}(x_0)$.*

Proof. Let A be a reversible d-dim net and let $x_0 \in \uparrow C^A_{\mathrm{suf}}$ be a nonlive marking of A. See Figure 3 for illustration. The value $C^A_{\mathrm{qd}} = f_{\mathrm{qd}}(||A||, d)$ will arise from the following analysis.

We fix a dead marking $y \in \mathrm{Reach}(x_0)$—it exists, since x_0 is nonlive—and a nonempty index set $I \subseteq [1, d]$ such that the restricted marking $y_{|I} \in [0, C^A_{\mathrm{dead}} - 1]^I$ is dead in the restricted net $A_{|I}$ (recall (3)). We have

$$y(i) < x_0(i) \text{ for every } i \in I, \tag{10}$$

because $y(i) < C^A_{\mathrm{dead}} \leq C^A_{\mathrm{suf}} \leq x_0(i)$ for every $i \in I$.

Since $x_0 \xrightarrow{*} y$, we obtain $y \overset{*}{\dashrightarrow} x_0$ by the reversibility of A. We then consider a segmented virtual execution

$$y = y_0 \overset{*}{\dashrightarrow} y_1 \overset{*}{\dashrightarrow} y_2 \cdots \overset{*}{\dashrightarrow} y_k = x_0 \tag{11}$$

as in (5). Hence, $||y_{j+1} - y_j|| \leq C^A_{\mathrm{shift}}$ by (6), and for each $i \in [1, d]$ the following holds:

- If $y(i) < x_0(i)$, then $y(i) < y_1(i) < y_2(i) \cdots < y_{j_i}(i) = y_{j_i+1}(i) \cdots = y_k(i) = x_0(i)$;
- If $y(i) \geq x_0(i)$, then $y_j(i) \geq x_0(i)$ for all $j \in [0, k]$.

We have thus highlighted the strict increase in the case $y(i) < x_0(i)$, and the maintenance of the threshold $x_0(i)$ in the case $y(i) \geq x_0(i)$.

Since $x_0(i) \geq C^A_{\mathrm{suf}}$ for all $i \in [1, d]$, there exists $j \leq C^A_{\mathrm{suf}}$ such that the marking y_j in (11) is above C^A_{suf}. By setting $x = y_j$, we can rewrite (11) as

$$y \overset{*}{\dashrightarrow} x \overset{*}{\dashrightarrow} x_0, \text{ where } x \in \uparrow C^A_{\mathrm{suf}} \text{ and } ||x - y|| \leq C^A_{\mathrm{suf}} \cdot C^A_{\mathrm{shift}}. \tag{12}$$

We have thus obtained a candidate for a quasi-dead marking $x \in \uparrow C^A_{\mathrm{suf}}$, since

$$x_0 \xrightarrow{*} x \xrightarrow{*} y. \tag{13}$$

Indeed, $x_0 \xrightarrow{*} y \overset{*}{\dashrightarrow} x$ implies $x_0 \overset{*}{\dashrightarrow} x$, hence $x_0 \xrightarrow{*} x$ by Lemma 3.4, since $x_0, x \in \uparrow C^A_{\mathrm{suf}}$ and $C^A_{\mathrm{suf}} = \max\{C^A_{\mathrm{dead}}, C^A_{\mathrm{vr}}\} \geq C^A_{\mathrm{vr}}$. Due to reversibility, we also have $x \overset{*}{\dashrightarrow} x_0$, and thus $x \xrightarrow{*} x_0$, since $x, x_0 \in \uparrow C^A_{\mathrm{vr}}$. Consequently, $x \xrightarrow{*} y$ (because $x \xrightarrow{*} x_0 \xrightarrow{*} y$).

However, we have not bounded the length of an execution from x to a dead marking. To handle this, we recall that our choice of the virtual execution $y \overset{*}{\dashrightarrow} x$ in (12) obtained from (11) guarantees that for each $i \in [1, d]$ we have

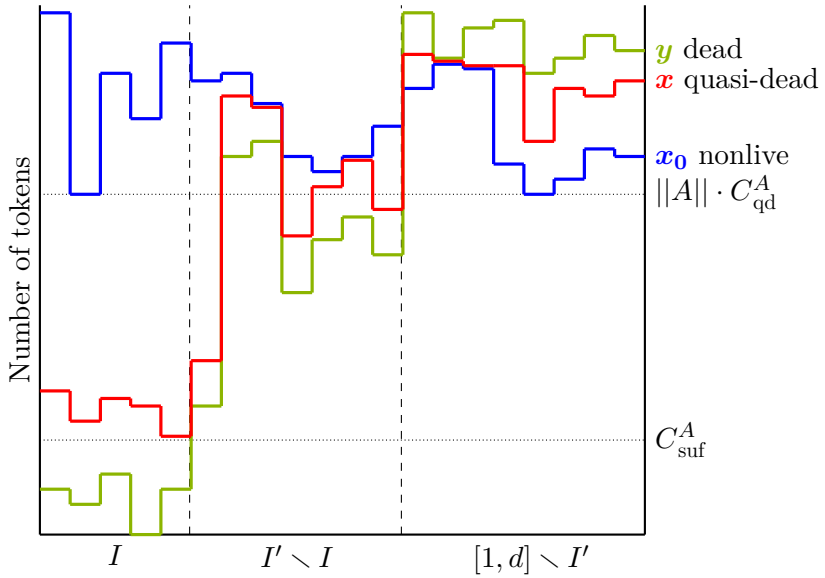


FIGURE 3. Illustration for the proof of Lemma 3.13.

$$\text{either } y(i) < x(i) \leq x_0(i) \text{ or } y(i) \geq x(i) \geq x_0(i).$$

As illustrated in Figure 3, we consider the set

$$I' = \{i \in [1,d] \mid y(i) < x(i)\} = \{i \in [1,d] \mid y(i) < x_0(i)\} \supseteq I;$$

the inclusion $I \subseteq I'$ follows from (10). Since $y_{|I}$ is dead in $A_{|I}$, the marking $y_{|I'}$ is dead in the net $A_{|I'}$ (by Proposition 3.35).

Because $x_{|I'} \geq y_{|I'}$ and $x_{|I'} \xrightarrow{*} y_{|I'}$ in $A_{|I'}$ (due to $x \xrightarrow{*} y$ in A), we can apply Proposition 3.12 to the net $A_{|I'}$. We obtain $x' \geq x_{|I'}$ (where $x' \in \mathbb{N}^{I'}$) satisfying:

a) $||x' - x_{|I'}|| \leq C^{A_{|I'}}_{\text{mr}} \cdot ||x_{|I'} - y_{|I'}|| \leq C^A_{\text{mr}} \cdot ||x - y|| \leq C^A_{\text{mr}} \cdot C^A_{\text{suf}} \cdot C^A_{\text{shift}}$ (using (12)); and

b) $x' \xrightarrow{\sigma} y_{|I'}$ for some σ such that $|\sigma| \leq C^{A_{|I'}}_{\text{mr}} \cdot ||x_{|I'} - y_{|I'}|| \leq C^A_{\text{mr}} \cdot C^A_{\text{suf}} \cdot C^A_{\text{shift}}$.

(Condition c) in Proposition 3.12 is irrelevant here.)

In $A_{|I'}$, we thus have $x_{|I'} \xrightarrow{*} y_{|I'}$ and $x' \xrightarrow{*} y_{|I'}$. By reversibility of $A_{|I'}$ (inherited from A, recall Proposition 2.3), we obtain $x_{|I'} \xdashrightarrow{*} x'$. Since $x' \geq x_{|I'} \in \uparrow C^A_{\text{suf}}$ and $C^A_{\text{suf}} \geq C^A_{\text{vr}} \geq C^{A_{|I'}}_{\text{vr}}$, Lemma 3.4 yields $x_{|I'} \xrightarrow{\sigma'} x'$ for a sequence σ' of actions in $A_{|I'}$, and thus in A by our conventions, such that

$$|\sigma'| \leq f_{\text{vr}}(||A_{|I'}||, |I'|) \cdot ||x' - x_{|I'}|| \leq C^A_{\text{vr}} \cdot C^A_{\text{mr}} \cdot C^A_{\text{suf}} \cdot C^A_{\text{shift}} \tag{14}$$

(using a)). To summarize, we have obtained the execution

$$x_{|I'} \xrightarrow{\sigma'} x' \xrightarrow{\sigma} y_{|I'} \text{ in } A_{|I'}, \tag{15}$$

where $y_{|I'}$ is a dead marking, $|\sigma'|$ is bounded in (14), and $|\sigma|$ is bounded in b). Hence, $|\sigma'\sigma| \leq f_{\mathrm{qd}}(||A||, d)$, if we set

$$f_{\mathrm{qd}}(||A||, d) = C^A_{\mathrm{qd}} = (C^A_{\mathrm{vr}} + 1) \cdot C^A_{\mathrm{mr}} \cdot C^A_{\mathrm{suf}} \cdot C^A_{\mathrm{shift}}. \tag{16}$$

Recall that $x(i) \geq x_0(i)$ for all $i \in [1, d] \smallsetminus I'$. Consequently, if the considered nonlive marking

$$x_0 \text{ is above the level } ||A|| \cdot C^A_{\mathrm{qd}},$$

we have $x(i) \geq ||A|| \cdot C^A_{\mathrm{qd}}$ for all $i \in [1, d] \smallsetminus I'$. Since $|\sigma'\sigma| \leq C^A_{\mathrm{qd}}$, the sequence $\sigma'\sigma$ can be executed not only from $x_{|I'}$ in $A_{|I'}$ (recall (15)) but also from x in A. We obtain an execution $x \xrightarrow{\sigma'\sigma} y'$ in A, where y' is a dead marking because $y'_{|I'} = y_{|I'}$ (and thus $y'_{|I} = y_{|I}$, where $y_{|I}$ is a dead marking in $A_{|I}$). The candidate x in (13) is thus indeed an f_{qd}-quasi-dead marking, for the 2-exp function f_{qd} defined by (16). □

Live solutions of $(\boldsymbol{A, x_0, \lambda})$-linear systems. Further, we fix a 2-exp function f_{qd} as guaranteed by Lemma 3.13. This fixes the constant $C^A_{\mathrm{qd}} = f_{\mathrm{qd}}(||A||, d)$, and thus the notion of quasi-dead marking, for every d-dim net A. The next lemma refers to Definition 3.8.

Lemma 3.14 (**(A, x_0, λ)-solutions are live, for sufficiently large live x_0 and λ**).
Let A be a reversible d-dim net, and let $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_d)$ be an extractor satisfying

- $\lambda_1 \geq C^A_{\mathrm{suf}} + ||A|| \cdot C^A_{\mathrm{qd}}$;
- $\lambda_{i+1} \geq \lambda_1 + ||A|| \cdot C^A_{\mathrm{vr}} \cdot (\lambda_i - C^A_{\mathrm{suf}})$ *for each* $i \in [1, d-1]$.

Let x_0 be a live marking of A satisfying

$$x_0 \in \uparrow \max\{C^A_{\mathrm{suf}}, ||A|| \cdot C^A_{\mathrm{qd}}\}.$$

Let S be an (A, x_0, λ)-linear system, and let $(\bar{x}_0, (\bar{x}_I)_{\emptyset \neq I \subseteq [1,d]})$ be a solution of S. Then $\bar{x}_0$ is a live marking.

Proof. Let A, λ, and x_0 be as in the statement. Let S be an (A, x_0, λ)-linear system (8), namely

$$(\mathbf{x} \equiv_B x_0) \wedge \bigwedge_{\emptyset \neq I \subseteq [1,d]} \big((\mathbf{x}_I \equiv_B x_I^{\mathrm{close}}) \wedge (\mathbf{x} \overset{*}{\dashrightarrow} \mathbf{x}_I)\big),$$

with $B = \max\{C^A_{\mathrm{suf}}, \lambda_d + C^A_{\mathrm{shift}}\}$. Recall that $(x_0, (x_I^{\mathrm{close}})_{\emptyset \neq I \subseteq [1,d]})$ is a solution of S, and that it is a tuple of live markings above C^A_{suf} (they are above C^A_{suf} by Definition 3.8, and live by Proposition 2.4, since x_0 is live and $x_I^{\mathrm{close}} \in \mathrm{Reach}(x_0)$). Towards a contradiction, suppose that

$$(\bar{x}_0, (\bar{x}_I)_{\emptyset \neq I \subseteq [1,d]}) \text{ is a solution of } S \text{ where } \bar{x}_0 \text{ is nonlive.}$$

Note that for each $\emptyset \neq I \subseteq [1, d]$, we have $\bar{x}_0 \overset{*}{\dashrightarrow} \bar{x}_I$ and $\bar{x}_I \equiv_B x_I^{\mathrm{close}}$, that is,

- $\bar{x}_I(i) = x_I^{\mathrm{close}}(i)$ for all $i \in [1, d]$ such that $x_I^{\mathrm{close}}(i) \leq B$, and
- $\bar{x}_I(i) > B$ for all $i \in [1, d]$ such that $x_I^{\mathrm{close}}(i) > B$.

Since $x_I^{\mathrm{close}} \in \uparrow C^A_{\mathrm{suf}}$ by definition, it follows that $\bar{x}_I \in \uparrow C^A_{\mathrm{suf}}$ as well. Moreover, the sets of indices of small components $I^{\lambda}_{\bar{x}_I}$ and $I^{\lambda}_{x_I^{\mathrm{close}}}$ (as defined in Proposition 3.6) are identical, and the markings $\bar{x}_I$ and x_I^{close} coincide on these indices.

Lemma 3.13 implies the existence of

$$\text{a quasi-dead marking } x \in \uparrow C^A_{\mathrm{suf}} \text{ such that } \bar{x}_0 \xrightarrow{*} x\text{; let } I = I^{\lambda}_x.$$

Since x is a quasi-dead marking, we have $x \xrightarrow{\sigma_1} y$ where y is a dead marking and $|\sigma_1| \leq C^A_{\mathrm{qd}}$. Note that $I \neq \emptyset$. Indeed, we have $y(i) < C^A_{\mathrm{dead}}$ for at least one $i \in [1,d]$ (recall (3)). The case $I = I^{\lambda}_x = \emptyset$ would imply, for all $i \in [1,d]$,

$$x(i) \geq \lambda_1 \geq C^A_{\mathrm{suf}} + ||A|| \cdot C^A_{\mathrm{qd}} \geq C^A_{\mathrm{dead}} + ||A|| \cdot |\sigma_1|;$$

this would in turn imply $y \in \uparrow C^A_{\mathrm{dead}}$—a contradiction. Consequently, $\bar{x}_I$ is well-defined.

Since all markings $\bar{x}_0, \bar{x}_I, x$ are above C^A_{suf} and A is reversible, we have

$$\bar{x}_0 \xrightarrow{*} \bar{x}_I \xrightarrow{*} \bar{x}_0 \xrightarrow{*} x \xrightarrow{*} \bar{x}_0\text{, hence } \bar{x}_I \xrightarrow{*} x \text{ in particular.}$$

Note that $(\lambda, I)\text{-dist}(\bar{x}_I) = (\lambda, I)\text{-dist}(x_I^{\mathrm{close}}) = 0$ (the distance to the λ-case I; recall Figure 2(right)). The equality $(\lambda, I)\text{-dist}(\bar{x}_I) = (\lambda, I)\text{-dist}(x_I^{\mathrm{close}})$ holds due to $\bar{x}_I \equiv_B x_I^{\mathrm{close}}$. If we had $(\lambda, I)\text{-dist}(\bar{x}_I) > 0$, then the first small segment $\bar{x}_I \xrightarrow{\sigma} x'$ of the virtual execution $\bar{x}_I \xrightarrow{*} x$ in the form of (5)—for which $||x' - \bar{x}_I|| \leq C^A_{\mathrm{shift}}$ and $x'(i)$ is between the values $\bar{x}_I(i)$ and $x(i)$ for each $i \in [1,d]$—would decrease this distance (that is, $(\lambda, I)\text{-dist}(x') < (\lambda, I)\text{-dist}(\bar{x}_I)$). For the sequence σ from $\bar{x}_I \xrightarrow{\sigma} x'$, we obtain $x_I^{\mathrm{close}} \xrightarrow{\sigma} x''$, where $x' \equiv_{\lambda_d} x''$ (since $\bar{x}_I \equiv_B x_I^{\mathrm{close}}$ and $B = \lambda_d + C^A_{\mathrm{shift}}$). Consequently, $x_I^{\mathrm{close}} \xrightarrow{*} x''$ (as both x_I^{close} and x'' are above $C^A_{\mathrm{suf}} \geq C^A_{\mathrm{vr}}$), and $(\lambda, I)\text{-dist}(x'') < (\lambda, I)\text{-dist}(x_I^{\mathrm{close}})$; this contradicts the choice of x_I^{close}. We thus have

$$(\lambda, I)\text{-dist}(x) = (\lambda, I)\text{-dist}(\bar{x}_I) = (\lambda, I)\text{-dist}(x_I^{\mathrm{close}}) = 0,$$

and all markings $x, \bar{x}_I, x_I^{\mathrm{close}}$ are above C^A_{suf}. Hence, for each $i \in I$, the values $x(i)$ and $\bar{x}_I(i) = x_I^{\mathrm{close}}(i)$ are in the interval $[C^A_{\mathrm{suf}}, \lambda_{|I|} - 1]$. The fact $\bar{x}_I \xrightarrow{*} x$ implies that in the restricted net $A_{|I}$ we have $(\bar{x}_I)_{|I} \xrightarrow{\sigma_2} x_{|I}$ where

$$|\sigma_2| \leq f_{\mathrm{vr}}(||A||, |I|) \cdot ||x_{|I} - (\bar{x}_I)_{|I}|| \leq C^A_{\mathrm{vr}} \cdot (\lambda_{|I|} - C^A_{\mathrm{suf}}).$$

Combining with $x \xrightarrow{\sigma_1} y$, where $|\sigma_1| \leq C^A_{\mathrm{qd}}$ and y is dead, we obtain

$$(\bar{x}_I)_{|I} = (x_I^{\mathrm{close}})_{|I} \xrightarrow{\sigma_2} x_{|I} \xrightarrow{\sigma_1} y_{|I}$$

in $A_{|I}$, where $|\sigma_2\sigma_1| \leq C^A_{\mathrm{vr}} \cdot (\lambda_{|I|} - C^A_{\mathrm{suf}}) + C^A_{\mathrm{qd}}$. Since for each $i \in [1,d] \smallsetminus I$, we have $x_I^{\mathrm{close}}(i) \geq \lambda_{|I|+1} \geq ||A|| \cdot |\sigma_2\sigma_1|$, it follows that

$$x_I^{\mathrm{close}} \xrightarrow{\sigma_2\sigma_1} y'$$

is an execution in A, where $y'_{|I} = y_{|I}$. We complete the proof by showing that y' is dead, which contradicts the assumption that x_I^{close} is live (since $x_0 \xrightarrow{*} x_I^{\mathrm{close}}$ and x_0 is live).

Recall that $x \xrightarrow{\sigma_1} y$ and $|\sigma_1| \leq C^A_{\mathrm{qd}}$. Since for each $i \in [1,d] \smallsetminus I$, we have

$$x(i) \geq \lambda_{|I|+1} \geq \lambda_1 \geq C^A_{\mathrm{dead}} + ||A|| \cdot C^A_{\mathrm{qd}},$$

it follows that $y(i) \geq C^A_{\mathrm{dead}}$ for each $i \in [1,d] \smallsetminus I$. This implies that $y_{|I}$ is dead in $A_{|I}$, which in turn implies that y' is dead in A (since $y'_{|I} = y_{|I}$). □

Petri nets with states (PNSs). Lemma 3.14, which gives conditions under which solutions of (A, x_0, λ)-linear systems are live, and Proposition 3.9, which guarantees 2-exp solutions of such systems, together establish Theorem 2.13 for reversible nets A that have a live marking x_0 above the level $\max\{C^A_{\mathrm{suf}}, ||A|| \cdot C^A_{\mathrm{qd}}\}$.

However, there exist structurally live conservative nets (hence, nets with live bottom SCCs) that have no live marking above that level. Indeed, there are nets in which the values of some components—indexed by a set $\emptyset \neq I \subseteq [1,d]$—remain small in every live marking, as demonstrated by Example 2.6. In this case, it is natural to view the restrictions $x_{|I}$ of (live) markings x as *control states*; the components (that is, Petri net places) outside I are viewed as *counters*. Definition 3.16 introduces *Petri nets with states* (PNSs), in a specific form tailored to our use. We will extend the previous results from nets to PNSs, with live markings having sufficiently large counter values.

Remark 3.15. Petri nets with states can be viewed, in principle, as *vector addition systems with states* (VASSs); a minor difference is that transitions between states are labelled with Petri net actions $a = (a_-, a_+)$ in PNSs, whereas in VASSs they are labelled merely with plain vectors encoding the effects $\Delta(a)$. Our definition of PNSs is also related to the notion of *rigid components* introduced in [Kos82] in order to keep track of the exact values of bounded components.

Given a d-dim net A and $I \subseteq [1,d]$, we say that a *nonempty finite* set $Q \subseteq \mathbb{N}^I$ is a *bSCC* of the restricted net $A_{|I}$ if it corresponds to a bottom SCC in the reachability graph of $A_{|I}$; that is, if in $A_{|I}$ we have $\mathrm{Reach}(x) = Q$ for all $x \in Q$. Figure 1 shows a 5-dim net A_1 and a restricted 4-dim net $A_2 = (A_1)_{|[1,4]}$. Recalling Example 2.6, we can verify that $Q = \{(1,0,1,0), (0,1,2,0), (0,1,0,1), (1,0,0,1)\}$ is a bSCC of A_2.

Definition 3.16 (PNS $G^A_{(I,Q)}$, norm $||G||$, relations $\xrightarrow{*}$ and $\overset{*}{\dashrightarrow}$, reversible PNS). Given a d-dim net A, an index set $I \subseteq [1,d]$, and a (finite) bSCC $Q \subseteq \mathbb{N}^I$, the *Petri net with states (PNS)* $G^A_{(I,Q)}$ is obtained from A by restricting *markings* $\bar{x} \in \mathbb{N}^d$ to those satisfying $\bar{x}_{|I} \in Q$. Setting $J = [1,d] \smallsetminus I$, such a marking $\bar{x} \in \mathbb{N}^d$ is also written as

$$(p, x) \in Q \times \mathbb{N}^J \text{ where } p = \bar{x}_{|I} \in Q \text{ and } x = \bar{x}_{|J} \in \mathbb{N}^J.$$

For $G = G^A_{(I,Q)}$, we say that G is of *dimension* d (inherited from A), or that G is a *d-dim PNS*. We define the *norm of* G as

$$||G|| = \max\{||Q||, ||A||\}.$$

The relations $\xrightarrow{a}$, $\xrightarrow{\sigma}$, $\xrightarrow{*}$ in $G^A_{(I,Q)}$ coincide with the respective relations in A restricted to markings in $Q \times \mathbb{N}^J$. The *virtual reachability relation* in $G^A_{(I,Q)}$ is defined on the set $Q \times \mathbb{Z}^J$ as follows: for action a in A we have

$$(p, x) \overset{a}{\dashrightarrow} (q, y) \text{ if } p \xrightarrow{a_{|I}} q \text{ in } A_{|I} \text{ and } x \overset{a_{|J}}{\dashrightarrow} y \text{ in } A_{|J}.$$

This naturally induces the relations $(p,x) \overset{\sigma}{\dashrightarrow} (q,y)$ for sequences σ of actions in A, and the virtual reachability relation $(p,x) \overset{*}{\dashrightarrow} (q,y)$ on the set $Q \times \mathbb{Z}^J$. Virtual reachability in a PNS $G^A_{(I,Q)}$ is thus based on the standard reachability within Q in $A_{|I}$:

$$(p, x) \overset{\sigma}{\dashrightarrow} (q, y) \text{ if and only if } p \xrightarrow{\sigma} q \text{ in } A_{|I} \text{ and } y - x = \Delta(\sigma)_{|J}.$$

(By our convention, we may write $p \xrightarrow{\sigma} q$ as a shorthand for $p \xrightarrow{\sigma_{|I}} q$.)

A *PNS* $G^A_{(I,Q)}$ is *reversible* if its virtual reachability relation $\overset{*}{\dashrightarrow}$ is symmetric; that is, $(p,x) \overset{*}{\dashrightarrow} (q,y)$ implies $(q,y) \overset{*}{\dashrightarrow} (p,x)$.

Remark 3.17. The dimension of $G = G^A_{(I,Q)}$, where A is a d-dim net and $I \subseteq [1,d]$, could be naturally defined as $|J|$ for $J = [1,d] \smallsetminus I$. This would correspond to the standard notion of dimension in vector addition systems with states (VASSs). However, we define it as d to emphasize that the underlying net A remains the primary object of study.

Note that the requirement that the state set Q in $G = G^A_{(I,Q)}$ is a bSCC of $A_{|I}$ guarantees that for any marking $\bar{x} = (p,x) \in Q \times \mathbb{N}^J$, the reachability set $\mathrm{Reach}(\bar{x})$ in A coincides with that in G.

Extractors $\boldsymbol{\lambda}$ yielding reversible PNSs $\boldsymbol{G^{(A,\lambda)}_{(I,Q)}}$ for live bottom SCCs of nets $\boldsymbol{A}$. The following lemma is a simplified variant of Lemma 20 in Section 6 of [Ler19]. It concerns a general SCC of a net; a bottom SCC is a special case. It allows us to separate, within an SCC, the "control states"—components whose values are small in every marking of the SCC—from the "counters", whose values are simultaneously large in at least one marking. The lemma uses the notion of $||A||$-*adapted extractors*. For later convenience, we also introduce the notion of *strictly* $||A||$-*adapted extractors*.

Definition 3.18 (**$||A||$-adapted and strictly $||A||$-adapted extractors $\boldsymbol{\lambda}$**).
Given a d-dim extractor $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_d)$, with $\lambda_0 = 1$ by our convention, and $m \in \mathbb{N}_+$ (in particular, $m = ||A||$ for a d-dim net A), we say that λ is *m-adapted* if for each $i \in [0, d-1]$,

$$\lambda_{i+1} \geq \lambda_i + m \cdot (\lambda_i)^i.$$

We say that λ is *strictly m-adapted* if $\lambda_1 \geq 1 + m$ and for each $i \in [1, d-1]$,

$$\lambda_{i+1} \geq m \cdot (\lambda_i)^i + m \cdot i \cdot (\lambda_i)^i. \tag{17}$$

(Note that every strictly m-adapted extractor is m-adapted.)

Lemma 3.19 [Ler19] (**Small control-state projection with large counters**).
Let X be an SCC in the reachability graph of a d-dim net A and let $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_d)$ be a d-dim extractor that is $||A||$-adapted. There exists a set $I \subseteq [1,d]$ satisfying the following conditions, where $J = [1,d] \smallsetminus I$:

a) $||X_{|I}|| < \lambda_{|I|}$ (hence for all $x \in X$ and $i \in I$ we have $x(i) < \lambda_{|I|}$);

b) there exists $x \in X$ such that for all $j \in J$ we have $x(j) \geq \lambda_{|I|+1} - ||A|| \cdot |I| \cdot (\lambda_{|I|})^{|I|}$.

In the previous lemma, the mere existence of a respective set $I \subset [1,d]$ is sufficient for our aims; we do not require its uniqueness in this case. This entails that we do not require the uniqueness of the PNS $G^{(A,\lambda)}_{(I,Q)}$ in the following definition.

Definition 3.20 (**Extracting PNS $\boldsymbol{G^{(A,\lambda)}_{(I,Q)}}$ from a live bottom SCC of a net $\boldsymbol{A}$**).
Let a (finite or infinite) set $X \subseteq \mathbb{N}^d$ be a live bottom SCC in the reachability graph of a d-dim net A. (The net A is reversible by Lemma 2.14.) Let λ be a strictly $||A||$-adapted d-dim extractor, and let $I \subseteq [1,d]$ be a set guaranteed by Lemma 3.19. We then view the PNS $G = G^A_{(I,Q)}$ with $Q = X_{|I}$ as a PNS $G^{(A,\lambda)}_{(I,Q)}$, *extracted from the SCC X via the extractor λ*.

In the previous definition, we consider strictly $||A||$-adapted extractors to guarantee that $Q = X_{|I}$ is a bSCC of $A_{|I}$ (as required in Definition 3.16):

Proposition 3.21 (**PNSs $\boldsymbol{G^{(A,\lambda)}_{(I,Q)}}$ are well-defined, and reversible**).
In every PNS $G^{(A,\lambda)}_{(I,Q)}$, the set Q is indeed a bSCC of $A_{|I}$. Moreover, $G^{(A,\lambda)}_{(I,Q)}$ is reversible.

Proof. Let a PNS $G = G^{(A,\lambda)}_{(I,Q)}$ be extracted from a live bottom SCC X of a d-dim net A as described in Definition 3.20; this implies that A is reversible, $Q = X_{|I}$, and $||Q|| < \lambda_{|I|}$ (by condition a) in Lemma 3.19). Consequently,

$$|Q| \leq (1 + ||Q||)^{|I|} \leq (\lambda_{|I|})^{|I|}.$$

If $I = \emptyset$, then $Q = \{()\}$ (the empty tuple is the unique control state), and we have $() \xrightarrow{a} ()$ for all actions in A. Hence, Q is a trivial bSCC of $A_{|I}$, and G is reversible because A is reversible. If $I = [1,d]$, then $A_{|I} = A$ and $Q = X$. Here, the facts that Q is a bSCC of $A_{|I}$ and that G is reversible follow from X being a bottom SCC of A. We thus further assume $\emptyset \neq I \neq [1,d]$.

To prove that $Q = X_{|I}$ is a bSCC of $A_{|I}$, we assume $p \xrightarrow{a} p'$ (that is, $p \xrightarrow{a_{|I}} p'$) in $A_{|I}$ with $p \in Q$, and show that $p' \in Q$; more precisely, we show that there exists a step $(p,y) \xrightarrow{a} (p',y')$ within X for some $y, y' \in \mathbb{N}^J$, where $J = [1,d] \smallsetminus I$. We use the fact that by condition b) in Lemma 3.19, there exists $(q_0, x_0) \in X$ such that ($q_0 \in Q$ and) for each $j \in J$,

$$x_0(j) \geq \lambda_{|I|+1} - ||A|| \cdot |I| \cdot (\lambda_{|I|})^{|I|} \geq ||A|| \cdot (\lambda_{|I|})^{|I|} \geq ||A|| \cdot |Q|.$$

(We use the property (17) of strict $||A||$-adapted extractors.) Since $Q = X_{|I}$, X is an SCC, and $q, p \in Q$, there exists an execution

$$q_0 \xrightarrow{a_1} q_1 \xrightarrow{a_2} q_2 \cdots \xrightarrow{a_k} q_k = p$$

in $A_{|I}$ with $k \leq |Q| - 1$. The corresponding execution

$$(q_0, x_0) \xrightarrow{a_1} (q_1, x_1) \xrightarrow{a_2} (q_2, x_2) \cdots \xrightarrow{a_k} (q_k, x_k) = (p, y)$$

is contained within X, because x_0 is above $||A|| \cdot |Q| \geq (k+1) \cdot ||A||$, and thus each x_i is above $(k+1-i) \cdot ||A|| \geq ||A||$. The step $(p,y) \xrightarrow{a} (p', y + \Delta(a)_{|J})$ is thus within X as well.

The proved fact that $p \xrightarrow{a} p'$ with $p \in Q$ implies a step $(p,y) \xrightarrow{a} (p',y')$ within X shows not only that Q is a bSCC of $A_{|I}$, but also that $G^{(A,\lambda)}_{(I,Q)}$ is reversible: If $(p,x) \dashrightarrow (p',x')$, that is, $p \xrightarrow{a_{|I}} p'$ and $x' - x = \Delta(a)_{|J}$, then within the SCC X we have $(p,y) \xrightarrow{a} (p',y')$ with $y' - y = x' - x = \Delta(a)_{|J}$; this implies the existence of an execution $(p',y') \xrightarrow{\sigma} (p,y)$ (within X), and thus $(p',x') \dashrightarrow (p,x)$. □

Adapting the proof of Theorem 2.13 from reversible nets to reversible PNSs. The issue that a d-dim net A with a live bottom SCC X might not possess a live marking above a sufficiently large level—specifically above $\max\{C^A_{\mathrm{suf}}, ||A|| \cdot C^A_{\mathrm{qd}}\}$ as stated in Lemma 3.14—is, in principle, resolved by extracting a PNS $G = G^{(A,\lambda)}_{(I,Q)}$ from X. For a suitably chosen 2-exp extractor λ, the norm $||G|| = \max\{||Q||, ||A||\}$ is (at most) 2-exp, and there exists a live marking (p_0, x_0) with x_0 above a sufficiently large level (by part b) of Lemma 3.19).

In what follows, we therefore adapt the arguments from the previous proof for reversible nets—which can be viewed as reversible PNSs with a single state (the empty tuple)—to the full case of reversible PNSs.

(Virtual) reachability, 1-exp $\boldsymbol{f_{\mathbf{vr}}}$ and $\boldsymbol{C^G_{\mathbf{vr}}}$, 2-exp $\boldsymbol{C^G_{\mathbf{suf}} = \max\{C^A_{\mathbf{dead}}, C^G_{\mathbf{vr}}\}}$. We now recall a variant of [Ler19, Lemma 5] in a more general form than in Lemma 3.4:

Lemma 3.22 (Virtual and standard reachability in reversible PNSs, $\boldsymbol{f_{\mathbf{vr}}}$ and $\boldsymbol{C^G_{\mathbf{vr}}}$). *There exists a 1-exp function f_{vr} such that for every reversible d-dim PNS G, the following conditions hold, where $C^G_{\mathrm{vr}} = f_{\mathrm{vr}}(||G||, d)$:*

(1) If $(p,x) \overset{*}{\dashrightarrow} (q,y)$*, then* $(p,x) \overset{\sigma}{\dashrightarrow} (q,y)$ *for some* σ *satisfying* $|\sigma| \leq C^G_{\mathrm{vr}} \cdot ||y-x||$.

(2) If $(p,x) \overset{*}{\dashrightarrow} (q,y)$ *and* $x, y \in \uparrow C^G_{\mathrm{vr}}$*, then* $(p,x) \overset{\sigma}{\to} (q,y)$ *with* $|\sigma| \leq C^G_{\mathrm{vr}} \cdot ||y-x||$.

Remark 3.23. The cited result in [Ler19] concerns a broader class of reversible PNSs than those considered here and is technically more detailed. As in other parts of this paper, we prioritize a simplified presentation that suffices for our needs.

Dead markings of a PNS $G = G^A_{(I,Q)}$ are the dead markings y of the net A satisfying $y_{|I} \in Q$. Therefore, we do not introduce a separate constant C^G_{dead}, but simply use C^A_{dead}. The sufficiently large 2-exp constant from (4) now becomes

$$C^G_{\mathrm{suf}} = \max\{C^A_{\mathrm{dead}}, C^G_{\mathrm{vr}}\}. \tag{18}$$

Virtual reachability in reversible PNSs via simple cycles in control-state units. Control-state units of PNSs can be naturally viewed as graphs, allowing us to use standard graph-theoretic terminology:

Definition 3.24 (Arc-labelled multigraph (Q,E) corresponding to a PNS $G^A_{(I,Q)}$). Given a d-dim PNS $G = G^A_{(I,Q)}$, by (Q,E) we denote the arc-labelled multigraph over the set of actions A, where $p \xrightarrow{a} q$ is an arc in E if $p \xrightarrow{a_{|I}} q$ holds in the net $A_{|I}$.

By a *path* in (Q,E), we mean a sequence $p_0 \xrightarrow{a_1} p_1 \cdots \xrightarrow{a_k} p_k$; it is a *cycle* if $p_k = p_0$, and a *simple cycle* if, moreover, $0 \leq i < j \leq k-1$ implies $p_i \neq p_j$. By the *effect* of a cycle $p \xrightarrow{\sigma} p$, we mean the vector $\Delta(\sigma)_{|J} \in \mathbb{Z}^J$, where $J = [1,d] \smallsetminus I$ (noting that $\Delta(\sigma)_{|I} = \mathbf{0}$).

The following proposition motivates focusing on "cyclic virtual reachability".

Proposition 3.25 (Virtual reachability in PNSs is based on control-unit cycles). *Given a reversible* d*-dim PNS* $G = G^A_{(I,Q)}$*, if* $(p,x) \overset{*}{\dashrightarrow} (q,y)$*, then there exists an action sequence* σ *with* $|\sigma| \leq (|Q|-1)$*—which implies* $||\Delta(\sigma)|| \leq (|Q|-1) \cdot ||A||$*—such that*

$$(p,x) \overset{\sigma}{\dashrightarrow} (q,x') \overset{*}{\dashrightarrow} (q,y),$$

where $x' = x + \Delta(\sigma)_{|J}$ *for* $J = [1,d] \smallsetminus I$.

Proof. If $(p,x) \overset{*}{\dashrightarrow} (q,y)$ in a reversible PNS $G = G^A_{(I,Q)}$, we may simply consider a shortest path $p \xrightarrow{\sigma} q$ in the corresponding graph (Q,E); we thus obtain $(p,x) \overset{\sigma}{\dashrightarrow} (q,x')$ with $|\sigma| \leq |Q|-1$. By reversibility of G, the relation $\overset{*}{\dashrightarrow}$ is symmetric, which implies $(q,x') \overset{*}{\dashrightarrow} (p,x)$; together with the assumption $(p,x) \overset{*}{\dashrightarrow} (q,y)$, this yields $(q,x') \overset{*}{\dashrightarrow} (q,y)$. □

We note that simple cycles constitute the building blocks of cyclic virtual executions:

Proposition 3.26 (Zero-effect all-state cycles, and simple cycles as basic blocks). *Given a reversible* d*-dim PNS* $G = G^A_{(I,Q)}$*, the following conditions hold:*

(1) There exists a cycle $p_0 \xrightarrow{\sigma_0} p_0$ *in the corresponding graph* (Q,E) *that visits all states in* Q *and has zero effect; that is, it satisfies not only* $\Delta(\sigma_0)_{|I} = \mathbf{0}$ *but also* $\Delta(\sigma_0)_{|J} = \mathbf{0}$ *for* $J = [1,d] \smallsetminus I$.

(2) For every state $p_0 \in Q$*, the effects of cycles* $p_0 \xrightarrow{\sigma} p_0$ *are precisely those vectors in* $\mathbb{Z}^J$ *that can be expressed as sums of the effects of simple cycles in* (Q,E).

Proof. (1) By definition, Q is a bottom SCC in $A_{|I}$. Consequently, there is a cycle $p \xrightarrow{\sigma} p$ in the graph (Q,E) that visits all states $q \in Q$; the cycle corresponds to a virtual execution $(p,\mathbf{0}) \dashrightarrow (p,\Delta(\sigma)_{|J})$. By the symmetry of the relation $\overset{*}{\dashrightarrow}$, there must also exist a virtual execution $(p,\Delta(\sigma)_{|J}) \dashrightarrow (p,\mathbf{0})$ for some sequence σ'. Therefore, $p \xrightarrow{\sigma\sigma'} p$ is the claimed zero-effect cycle visiting all states.

(2) Let $\mathcal{E} \subseteq \mathbb{Z}^J$ be the set of effects of simple cycles in the graph (Q,E). Any cycle $p_0 \xrightarrow{\sigma} p_0$ in (Q,E) is either a simple cycle—in which case $\Delta(\sigma)_{|J} \in \mathcal{E}$—or it can be decomposed as $p_0 \xrightarrow{\sigma_1} q \xrightarrow{\sigma_2} q \xrightarrow{\sigma_3} p_0$, where $q \xrightarrow{\sigma_2} q$ is a simple cycle. In the latter case, the effect of the cycle $p_0 \xrightarrow{\sigma_1\sigma_3} p_0$ belongs to $\mathcal{E}^*$ by the inductive hypothesis; since $\Delta(\sigma_2)_{|J} \in \mathcal{E}$, it follows that $\Delta(\sigma_1\sigma_2\sigma_3)_{|J}$ belongs to $\mathcal{E}^*$ as well.

On the other hand, if $z_0 \in \mathcal{E}$—due to a simple cycle $q \xrightarrow{\sigma} q$ with $\Delta(\sigma)_{|J} = z_0$— then for any $p_0 \in Q$, we may consider a cycle $p_0 \xrightarrow{\sigma_1} q \xrightarrow{\sigma} q \xrightarrow{\sigma_2} p_0$, where $p_0 \xrightarrow{\sigma_1\sigma_2} p_0$ is a zero-effect cycle visiting all states in Q. Consequently, the effect of the cycle $p_0 \xrightarrow{\sigma_1\sigma\sigma_2} p_0$ is z_0. This implies that for any $z \in \mathcal{E}^*$, there exists a cycle $p_0 \xrightarrow{\sigma} p_0$ with the effect $\Delta(\sigma)_{|J} = z$. □

Cyclic virtual reachability in a PNS $\boldsymbol{G}$ via a "simple-cycle" net $\boldsymbol{A^G_{\mathrm{SC}}}$. Recall that a d-dim net A is a finite set of actions $a = (a_-, a_+) \in \mathbb{N}^d \times \mathbb{N}^d$, with $\Delta(a) = (a_+ - a_-) \in \mathbb{Z}^d$. For the virtual reachability relation $\overset{*}{\dashrightarrow}$, only the set $A_\delta = \{\Delta(a) \mid a \in A\}$ is relevant, since $x \overset{*}{\dashrightarrow} y$ holds if and only if $(y-x) \in A^*_\delta$. In the following definition, for $z \in \mathbb{Z}^J$ we use the notation z_+, z_- where, for each $i \in J$,

$$z_+(i) = \max\{z(i), 0\} \text{ and } z_-(i) = \max\{-z(i), 0\}.$$

For instance, $(2,-3,0)_+ = (2,0,0)$ and $(2,-3,0)_- = (0,3,0)$. Thus $z = z_+ - z_-$, and each $z \in \mathbb{Z}^J$ can be implicitly viewed as a net action $z = (z_+, z_-)$ with $\Delta(z) = z$.

Definition 3.27 (**Simple-cycle $|J|$-dim net A^G_{SC} associated with a PNS $G = G^A_{(I,Q)}$**). Given a d-dim PNS $G = G^A_{(I,Q)}$ with the corresponding graph (Q,E) and $J = [1,d] \setminus I$, the *simple-cycle net* A^G_{SC} is the $|J|$-dim net

$$A^G_{\mathrm{SC}} = (A^G_{\mathrm{SC}})_\delta = \{z \in \mathbb{Z}^J \mid z \text{ is the effect of a simple cycle in } (Q,E)\}.$$

We note that, given a d-dim PNS $G = G^A_{(I,Q)}$, we have

$$||A^G_{\mathrm{SC}}|| \leq |Q| \cdot ||A|| \leq ||A|| \cdot (1 + ||Q||)^{|I|}. \tag{19}$$

The following fact is a corollary of Proposition 3.26(2).

Corollary 3.28 (**Reducing cyclic virtual reachability in G to the net A^G_{SC}**).
Given a reversible d-dim PNS $G = G^A_{(I,Q)}$, for any $p \in Q$ we have

$$(p,x) \overset{*}{\dashrightarrow} (p,y) \text{ in } G \text{ if and only if } x \overset{*}{\dashrightarrow} y \text{ in } A^G_{\mathrm{SC}}.$$

Extending $(\boldsymbol{A, x_0, \lambda})$-linear systems to PNSs. We now extend the notion of (A, x_0, λ)-linear systems from Definition 3.8 to PNSs.

Definition 3.29 (**$((G,(p_0,x_0),\lambda)$-linear systems for reversible PNSs**).
Given a reversible d-dim PNS $G = G^A_{(I,Q)}$, where $J = [1,d] \setminus I$, and a $|J|$-dim extractor $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_{|J|})$, we define the bound

$$B = B_{(G,\lambda)} = \max\{C^G_{\mathrm{suf}}, \lambda_{|J|} + C^{A^G_{\mathrm{SC}}}_{\mathrm{shift}}\}. \tag{20}$$

Given a marking $(p_0, x_0) \in Q \times \mathbb{N}^J$ with $x_0 \in \uparrow C^G_{\mathrm{suf}}$, for each $\emptyset \neq J' \subseteq J$ let

$$x^{\mathrm{close}}_{J'} \in \mathbb{N}^J \cap \uparrow C^G_{\mathrm{suf}}$$

be a fixed vector such that $(p_0, x_0) \xrightarrow{*} (p_0, x^{\mathrm{close}}_{J'})$ and (λ, J')-dist$(x^{\mathrm{close}}_{J'})$ is minimal. Then the following linear system S of dimension $|J| \cdot 2^{|J|}$, with a $|J|$-dim variable $\mathbf{x} = (\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_{|J|})$ and $2^{|J|} - 1$ $|J|$-dim variables $\mathbf{x}_{J'}$, one for each $\emptyset \neq J' \subseteq J$, is

$$\text{a } (G, (p_0, x_0), \lambda)\text{-}\textit{linear system}\text{:}$$

$$(\mathbf{x} \equiv_B x_0) \wedge \bigwedge_{\emptyset \neq J' \subseteq J} \left((\mathbf{x}_{J'} \equiv_B x^{\mathrm{close}}_{J'}) \wedge (\mathbf{x} \xrightarrow{*} \mathbf{x}_{J'}) \right), \tag{21}$$

where $\mathbf{x} \xrightarrow{*} \mathbf{x}_{J'}$ refers to the virtual reachability in A^G_{SC}.

Note that (x, x') is a solution of the expression $\mathbf{x} \xrightarrow{*} \mathbf{x}_{J'}$ referring to A^G_{SC} if and only if $(p_0, x) \xrightarrow{*} (p_0, x')$ holds in G (due to Corollary 3.28).

Our aim is to show that if (p_0, x_0) is a live marking of $G^A_{(I,Q)}$, where x_0 is above a sufficiently large level relative to $||G|| = \max\{||Q||, ||A||\}$, then $(p_0, \bar{x}_0)$ is live for every solution $(\bar{x}_0, (\bar{x}_{J'})_{\emptyset \neq J' \subseteq J})$ of a $(G, (p_0, x_0), \lambda)$-linear system, for a suitable 2-exp extractor λ. As previously, we use the notion of quasi-dead markings to this aim.

Definition 3.30 (**Quasi-dead markings in PNSs**).
For a 2-exp function f_{qd} and a d-dim PNS $G = G^A_{(I,Q)}$, where $J = [1, d] \smallsetminus I$, a marking $(p_0, x) \in Q \times \mathbb{N}^J$ is f_{qd}-*quasi-dead* in G, or simply *quasi-dead* in G when f_{qd} is clear from context, if $x \in \uparrow C^G_{\mathrm{suf}}$ and there exists an execution $(p_0, x) \xrightarrow{\sigma} (q, y)$ such that (q, y) is a dead marking and $|\sigma| \leq f_{\mathrm{qd}}(||G||, d) = C^G_{\mathrm{qd}}$.

Lemma 3.31 (**Nonlive (p_0, x_0) with large x_0 reaches a quasi-dead marking (p_0, x)**). *There exists a 2-exp function f_{qd} such that for every reversible d-dim PNS $G = G^A_{(I,Q)}$ and every nonlive marking $(p_0, x_0) \in Q \times \mathbb{N}^J$, where $J = [1, d] \smallsetminus I$ and*

$$x_0 \in \uparrow \left(2 \cdot (|Q| - 1) \cdot ||A|| + \max\{C^G_{\mathrm{suf}}, ||A|| \cdot C^G_{\mathrm{qd}}\} \right), \tag{22}$$

there exists an f_{qd}-quasi-dead marking $(p_0, x) \in \mathrm{Reach}((p_0, x_0))$.

Proof. Let $G = G^A_{(I,Q)}$ be a reversible d-dim PNS, where $J = [1, d] \smallsetminus I$, and let (p_0, x_0) be a nonlive marking with $x_0 \in \mathbb{N}^J$ satisfying

$$x_0 \in \uparrow \left(2 \cdot (|Q| - 1) \cdot ||A|| + C^G_{\mathrm{suf}} \right). \tag{23}$$

The value $C^G_{\mathrm{qd}} = f_{\mathrm{qd}}(||G||, d)$ will arise from the following analysis.

Recall that Q is a bSCC in $A_{|I}$. If there exists $p \in Q$ that is nonlive in $A_{|I}$—that is, there exists an action $a \in A$ that is not enabled at any $q \in Q$ in $A_{|I}$—then all $q \in Q$ are dead markings in $A_{|I}$ (by our terminology); consequently, (p_0, x_0) is dead in the net A, and thus in the PNS G as well. We thus further assume that Q is a live bottom SCC in $A_{|I}$.

We now fix a dead marking (q, y) such that $(p_0, x_0) \xrightarrow{*} (q, y)$. Let $\emptyset \neq J' \subseteq J$ be such that $y(i) < C^A_{\mathrm{dead}}$ for all $i \in J'$ and

$$(q, y_{|J'}) \text{ is dead in the net } A_{|(I \cup J')}. \tag{24}$$

By reversibility of G and the assumption (23), we may fix an execution

$$(p_0, x_0) \xrightarrow{\sigma_0} (q, y') \xrightarrow{*} (q, y)$$

where $p_0 \xrightarrow{\sigma_0} q$ is a shortest path from p_0 to q in the graph (Q, E) associated with G. (Recall Proposition 3.25, Lemma 3.22, and the fact that $C^G_{\mathrm{suf}} \geq C^G_{\mathrm{vr}}$ by (18).) Since $|\sigma_0| \leq |Q| - 1$, we obtain that, for all $i \in J$,

$$y'(i) \geq x_0(i) - (|Q| - 1) \cdot ||A||, \tag{25}$$

and thus

$$y' \in \uparrow \left((|Q| - 1) \cdot ||A|| + C^G_{\mathrm{suf}} \right). \tag{26}$$

Since $y' \overset{*}{\dashrightarrow} y$ in the $|J|$-dim "simple-cycle" net A^G_{SC} associated with G (by Corollary 3.28), due to the reversibility of G—implying the reversibility of A^G_{SC}—we may consider a segmented virtual execution

$$y = y_0 \overset{*}{\dashrightarrow} y_1 \overset{*}{\dashrightarrow} y_2 \cdots \overset{*}{\dashrightarrow} y_k = y' \tag{27}$$

as in (5), for the net A^G_{SC}. Consequently, for all $j \in [0, k-1]$ we have

$$||y_{j+1} - y_j|| \leq C^{A^G_{\mathrm{SC}}}_{\mathrm{shift}} = f_{\mathrm{shift}}(||A^G_{\mathrm{SC}}||, |J|). \tag{28}$$

Moreover, the following holds for each $i \in J$:

- If $y(i) < y'(i)$, then $y(i) < y_1(i) < y_2(i) \cdots < y_{j_i}(i) = y_{j_i+1}(i) \cdots = y_k(i) = y'(i)$;
- If $y(i) \geq y'(i)$, then $y_j(i) \geq y'(i)$ for all $j \in [0, k]$.

By (26), there exists $j \leq C^G_{\mathrm{suf}} + (|Q| - 1) \cdot ||A||$ such that the marking y_j in (27) is above the level $C^G_{\mathrm{suf}} + (|Q| - 1) \cdot ||A||$. Setting $x' = y_j$, we obtain

$$(q, y) \overset{*}{\dashrightarrow} (q, x') \xrightarrow{\sigma'} (p_0, x),$$

where $q \xrightarrow{\sigma'} p_0$ is a shortest path from q to p_0 in the graph (Q, E). Consequently, $|\sigma'| \leq |Q| - 1$, and x', x are both above C^G_{suf}, because

$$||x' - x|| \leq (|Q| - 1) \cdot ||A||. \tag{29}$$

Since $(p_0, x_0) \xrightarrow{*} (q, y) \overset{*}{\dashrightarrow} (q, x') \xrightarrow{\sigma'} (p_0, x)$ and all x_0, x', x are above $C^G_{\mathrm{suf}} \geq C^G_{\mathrm{vr}}$, we obtain

$$(p_0, x_0) \xrightarrow{*} (q, x') \xrightarrow{*} (p_0, x) \xrightarrow{*} (p_0, x_0),$$

where the last relation follows from reversibility. Since $(p_0, x_0) \xrightarrow{*} (q, y)$, we obtain

$$(p_0, x_0) \xrightarrow{*} (p_0, x) \xrightarrow{\sigma_1} (q, x') \xrightarrow{\sigma_2} (q, y)$$

for some action sequences σ_1, σ_2. The marking (p_0, x) is a candidate for a quasi-dead marking with the control state p_0. We can bound the length of σ_1, using Lemma 3.22 and (29):

$$|\sigma_1| \leq C^G_{\mathrm{vr}} \cdot ||x' - x|| \leq C^G_{\mathrm{vr}} \cdot (|Q| - 1) \cdot ||A||. \tag{30}$$

However, we have no bound on $|\sigma_2|$. To handle this, we proceed as in the proof of Lemma 3.13. We set

$$J'' = \{i \in J \mid y(i) < y'(i)\} = \{i \in J \mid y(i) < x'(i)\}, \tag{31}$$

noting that $J' \subseteq J'' \subseteq J$; recall (24) and the fact that $y(i) < C^A_{\mathrm{dead}} \leq C^G_{\mathrm{suf}} \leq y'(i)$ for all $i \in J'$. Since the marking $(q, y_{|J'})$ is dead in the net $A_{|(I\cup J')}$, the marking $(q, y_{|J''})$ is dead in the net $A_{|(I\cup J'')}$.

Since $(q, x'_{|J''}) \xrightarrow{*} (q, y_{|J''})$ in $A_{|(I\cup J'')}$, and $(q, x'_{|J''}) \geq (q, y_{|J''})$, we may apply Proposition 3.12. Consequently, there exists $x'' \in \mathbb{N}^{J''}$ such that $x'' \geq x'_{|J''}$ and $(q, x'') \xrightarrow{\sigma} (q, y_{|J''})$ for some sequence σ, where both $||x'' - x'_{|J''}||$ and $|\sigma|$ are bounded by

$$f_{\mathrm{mr}}(||A||, d) \cdot ||x' - y|| \leq C^A_{\mathrm{mr}} \cdot (C^G_{\mathrm{suf}} + (|Q| - 1) \cdot ||A||) \cdot C^{A^G_{\mathrm{sc}}}_{\mathrm{shift}}. \tag{32}$$

Using reversibility, we obtain $(q, x'_{|J''}) \xdashrightarrow{*} (q, x'')$ in the net $A' = A_{|(I\cup J'')}$, as well as in the PNS $G' = G_{|(I\cup J'')} = G^{A'}_{(I,Q)}$. Since $x'_{|J''}$ and x'' are above $C^G_{\mathrm{suf}} \geq C^G_{\mathrm{vr}} \geq C^{G'}_{\mathrm{vr}}$, there exists an action sequence ρ satisfying $(q, x'_{|J''}) \xrightarrow{\rho} (q, x'')$ and

$$|\rho| \leq C^G_{\mathrm{vr}} \cdot ||x'' - x'_{|J''}|| \leq C^G_{\mathrm{vr}} \cdot f_{\mathrm{mr}}(||A||, d) \cdot ||x' - y||. \tag{33}$$

We have thus obtained that, in the net $A_{|(I\cup J'')}$,

$$(p_0, x_{|J''}) \xrightarrow{\sigma_1} (q, x'_{|J''}) \xrightarrow{\rho} (q, x'') \xrightarrow{\sigma} (q, y_{|J''}).$$

Inspecting the bound on the length $|\sigma_1\rho\sigma|$ provided by (30), (33), (32), (28), and (19), we deduce the existence of a 2-exp function f_{qd} such that

$$(p_0, x_{|J''}) \xrightarrow{\sigma_1\rho\sigma} (q, y_{|J''}) \text{ with } |\sigma_1\rho\sigma| \leq f_{\mathrm{qd}}(||G||, d).$$

We have $(p_0, x) \xrightarrow{\sigma_1} (q, x')$ in G; however, the sequence $\rho\sigma$ might not be executable from (q, x') in G. Recalling (31), (25), and the definition $x' = y_j$ for a respective y_j from (27), we note that, for all $i \in J \setminus J''$,

$$x'(i) \geq y'(i) \geq x_0(i) - (|Q| - 1) \cdot ||A||.$$

Consequently, if x_0 (satisfying (23)) is above the level $(|Q| - 1) \cdot ||A|| + ||A|| \cdot C^G_{\mathrm{qd}}$, then we obtain

$$(p_0, x) \xrightarrow{\sigma_1} (q, x') \xrightarrow{\rho\sigma} (q, y'')$$

in G, where $y''_{|J''} = y_{|J''}$. Since $(q, y_{|J''})$ is dead in $A_{|(I\cup J'')}$, the marking (q, y'') is dead in A (and thus in G). Consequently, (p_0, x) is an f_{qd}-quasi-dead marking in G. □

Guaranteed existence of live markings $(\boldsymbol{p_0, x_0})$ with large counter values $\boldsymbol{x_0}$. We now show that from a live bottom SCC X of a d-dim net A, we can always extract a PNS $G = G^A_{(I,Q)}$ where $||G|| = \max\{||Q||, ||A||\}$ is bounded by a 2-exp function and which possesses a live marking (p_0, x_0) such that x_0 satisfies condition (22). To this end, we fix a 2-exp function f_{qd} provided by Lemma 3.31. By f_{suf}, we mean a 2-exp function satisfying $f_{\mathrm{suf}}(||G||, d) = C^G_{\mathrm{suf}} = \max\{C^A_{\mathrm{dead}}, C^G_{\mathrm{vr}}\}$ for every d-dim PNS $G^A_{(I,Q)}$ (which coincides with A if $I = \emptyset$).

Definition 3.32 (Feasible PNS $\boldsymbol{G^{(A,\lambda')}_{(I,Q)}}$ extracted from a live bottom SCC $\boldsymbol{X}$). Let a partial function $\bar{\lambda}\colon (\mathbb{N}_+)^3 \to \mathbb{N}_+$ be defined inductively as follows for all triples $(m, d, i) \in (\mathbb{N}_+)^3$ with $i \in [1, d]$; we write $\bar{\lambda}^{(m,d)}_i$ to denote $\bar{\lambda}(m, d, i)$:

- $\bar{\lambda}_1^{(m,d)} = 1 + \max\{f_{\text{suf}}(m,d), m \cdot f_{\text{qd}}(m,d)\}$;
- $\bar{\lambda}_{i+1}^{(m,d)} = \left[2 \cdot ((\bar{\lambda}_i^{(m,d)})^i - 1) \cdot m + \max\{f_{\text{suf}}(\bar{\lambda}_i^{(m,d)}, d), m \cdot f_{\text{qd}}(\bar{\lambda}_i^{(m,d)}, d)\}\right] + m \cdot i \cdot (\bar{\lambda}_i^{(m,d)})^i$.

For any live bottom SCC X in the reachability graph of a d-dim net A, a d-dim PNS $G = G^A_{(I,Q)}$ is *feasible* if $G = G^{(A,\lambda')}_{(I,Q)}$, where $\lambda' = (\lambda'_1, \lambda'_2, \ldots, \lambda'_d)$ is the extractor satisfying $\lambda'_i = \bar{\lambda}_i^{(||A||,d)}$ for all $i \in [1,d]$.
(Recall Definition 3.20 and note that λ' is strictly $||A||$-adapted.)

Observe that the function $\bar{f}(m,d) = \bar{\lambda}_d^{(m,d)}$ is a 2-exp function ($\bar{f}(m,d) \leq (2+m)^{2^{\text{poly}(d)}}$).

Proposition 3.33 (**Feasible PNSs possess live markings with large counter values**). *Let X be a live bottom SCC of a d-dim net A, and let $G = G^{(A,\lambda')}_{(I,Q)}$ be a feasible PNS extracted from X. Then there exists a (live) marking $(p_0, x_0) \in X$ with $p_0 \in Q = X_{|I}$ and*

$$x_0 \in \uparrow \left(2 \cdot (|Q| - 1) \cdot ||A|| + \max\{C^G_{\text{suf}}, ||A|| \cdot C^G_{\text{qd}}\}\right).$$

Proof. Let $A, X, G = G^{(A,\lambda')}_{(I,Q)}$ be as in the statement.

If $I = [1,d]$, then $Q = X$, and the claim holds trivially. If $I = \emptyset$, then the PNS G coincides with the net A (in this case, $Q = \{()\}$ and thus $|Q| = 1$), and, by definition of $G^{(A,\lambda')}_{(I,Q)}$, there exists $x_0 \in X$ such that, for all $i \in [1,d]$,

$$x_0(i) \geq \lambda'_1 \geq 1 + \max\{f_{\text{suf}}(||A||, d), ||A|| \cdot f_{\text{qd}}(||A||, d)\} \geq \max\{C^A_{\text{suf}}, ||A|| \cdot C^A_{\text{qd}}\}.$$

We therefore further assume $\emptyset \neq I \neq [1,d]$. By definition of $G = G^{(A,\lambda')}_{(I,Q)}$, we have $||Q|| < \lambda'_{|I|}$, and there exists $(p_0, x_0) \in X$ where, for every $i \in J = [1,d] \setminus I$,

$$x_0(i) \geq \lambda'_{|I|+1} - ||A|| \cdot |I| \cdot (\lambda'_{|I|})^{|I|}.$$

The definition of λ' (in Definition 3.32) implies that, for every $i \in J = [1,d] \setminus I$,

$$x_0(i) \geq 2 \cdot ((\lambda'_{|I|})^{|I|} - 1) \cdot ||A|| + \max\{f_{\text{suf}}(\lambda'_{|I|}, d), ||A|| \cdot f_{\text{qd}}(\lambda'_{|I|}, d)\}.$$

Since $(\lambda'_{|I|})^{|I|} - 1 \geq (1 + ||Q||)^{|I|} - 1 \geq |Q| - 1$ and $\lambda'_{|I|} \geq \max\{||Q||, ||A||\} = ||G||$, we obtain, for every $i \in J = [1,d] \setminus I$,

$$x_0(i) \geq 2 \cdot (|Q| - 1) \cdot ||A|| + \max\{f_{\text{suf}}(||G||, d), ||A|| \cdot f_{\text{qd}}(||G||, d)\},$$

that is,

$$x_0(i) \geq 2 \cdot (|Q| - 1) \cdot ||A|| + \max\{C^G_{\text{suf}}, ||A|| \cdot C^G_{\text{qd}}\}.$$ □

Live solutions of $(G, (p_0, x_0), \lambda)$-linear systems. Recall Definition 3.29 of $(G, (p_0, x_0), \lambda)$-linear systems. We now show that, given a feasible PNS $G = G^{(A,\lambda')}_{(I,Q)}$, if (p_0, x_0) is a live marking of G guaranteed by Proposition 3.33 and S is a $(G, (p_0, x_0), \lambda)$-linear system for an extractor λ analogous to that in Lemma 3.14, then $(p_0, \bar{x}_0)$ is a live marking of G for every solution $(\bar{x}_0, (\bar{x}_{J'})_{\emptyset \neq J' \subseteq J})$ of S. The proof is analogous to that of Lemma 3.14 and is therefore presented here in a more concise manner.

Lemma 3.34 (**$((G,(p_0,x_0),\lambda)$-solutions are live, if G is feasible and x_0,λ are large**). *Let $G = G^{(A,\lambda')}_{(I,Q)}$ be a feasible PNS, extracted from a live bottom SCC X of a d-dim net A. Let $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_{|J|})$, where $J = [1,d] \smallsetminus I$, be an extractor satisfying:*

- $\lambda_1 \geq C^G_{\mathrm{suf}} + ||A|| \cdot C^G_{\mathrm{qd}}$;
- $\lambda_{i+1} \geq \lambda_1 + ||A|| \cdot C^G_{\mathrm{vr}} \cdot (\lambda_i - C^G_{\mathrm{suf}})$ *for each* $i \in [1, |J|-1]$.

Let (p_0, x_0) be a live marking of G where $x_0 \in \mathbb{N}^J$ satisfies

$$x_0 \in \uparrow \left(2 \cdot (|Q|-1) \cdot ||A|| + \max\{C^G_{\mathrm{suf}}, ||A|| \cdot C^G_{\mathrm{qd}}\}\right).$$

Let S be a $(G,(p_0,x_0),\lambda)$-linear system, and let $(\bar{x}_0, (\bar{x}_{J'})_{\emptyset \neq J' \subseteq J})$ be a solution of S. Then $(p_0, \bar{x}_0)$ is a live marking.

Proof. Let $A, G = G^{(A,\lambda')}_{(I,Q)}, \lambda$, and (p_0,x_0) be as in the statement. Let $A' = A^G_{\mathrm{SC}}$ be the "simple-cycle" $|J|$-dim net associated with G, where $J = [1,d] \smallsetminus I$.

Let S be a $(G,(p_0,x_0),\lambda)$-linear system (21), namely

$$(\mathbf{x} \equiv_B x_0) \wedge \bigwedge_{\emptyset \neq J' \subseteq J} \left((\mathbf{x}_{J'} \equiv_B x^{\mathrm{close}}_{J'}) \wedge (\mathbf{x} \overset{*}{\dashrightarrow} \mathbf{x}_{J'})\right),$$

where $B = \max\{C^G_{\mathrm{suf}}, \lambda_{|J|} + C^{A'}_{\mathrm{shift}}\}$ and $\mathbf{x} \overset{*}{\dashrightarrow} \mathbf{x}_{J'}$ refers to the virtual reachability in A'. Recall that $(p_0,x_0) \overset{*}{\to} (p_0, x^{\mathrm{close}}_{J'})$ for each $\emptyset \neq J' \subseteq J$; this implies $x_0 \overset{*}{\dashrightarrow} x^{\mathrm{close}}_{J'}$ in A'.

Towards a contradiction, suppose that

$$(\bar{x}_0, (\bar{x}_{J'})_{\emptyset \neq J' \subseteq J}) \text{ is a solution of } S \text{ where } (p_0, \bar{x}_0) \text{ is nonlive.}$$

The sets of indices of λ-small components $I^{\lambda}_{\bar{x}_{J'}} \subseteq J$ and $I^{\lambda}_{x^{\mathrm{close}}_{J'}} \subseteq J$ (as defined in Proposition 3.6) are identical, and the vectors $\bar{x}_{J'}$ and $x^{\mathrm{close}}_{J'}$—viewed as markings of the net $A' = A^G_{\mathrm{SC}}$—coincide on these indices. By definition, all vectors $x_0, x^{\mathrm{close}}_{J'}$, and therefore also $\bar{x}_0, \bar{x}_{J'}$, are above the level C^G_{suf}. Lemma 3.31 implies the existence of

$$\text{a quasi-dead marking } (p_0,x) \in \mathrm{Reach}((p_0,\bar{x}_0)) \text{ with } x \in \uparrow C^G_{\mathrm{suf}}\text{; let } I' = I^{\lambda}_x \subseteq J.$$

Since (p_0,x) is a quasi-dead marking, we have $(p_0,x) \xrightarrow{\sigma_1} (q,y)$ where (q,y) is a dead marking and $|\sigma_1| \leq C^G_{\mathrm{qd}}$. Note that $I' \neq \emptyset$. Indeed, we have $y(i) < C^A_{\mathrm{dead}}$ for at least one $i \in J$, since otherwise q would be dead in $A_{|I}$, which would imply that (p_0,x_0) is nonlive (because $(p_0,x_0) \xrightarrow{\sigma_1} (q, x_0 + \Delta(\sigma_1)_{|J})$). The case $I' = I^{\lambda}_x = \emptyset$ would imply that x is above the level $\lambda_1 \geq C^G_{\mathrm{suf}} + ||A|| \cdot C^G_{\mathrm{qd}} \geq C^G_{\mathrm{suf}} + ||A|| \cdot |\sigma_1|$, in which case y would be above $C^G_{\mathrm{suf}} \geq C^A_{\mathrm{dead}}$—a contradiction. Consequently, $\bar{x}_{I'}$ is well-defined.

Since $(p_0,\bar{x}_0) \overset{*}{\to} (p_0,x)$, we have $\bar{x}_0 \overset{*}{\dashrightarrow} x$ in A'. By reversibility of A' (both A and G are reversible), it follows that $\bar{x}_{I'} \overset{*}{\dashrightarrow} x$ in A' (since $\bar{x}_0 \overset{*}{\dashrightarrow} \bar{x}_{I'}$ and $\bar{x}_0 \overset{*}{\dashrightarrow} x$); this implies $(p_0,\bar{x}_{I'}) \overset{*}{\to} (p_0,x)$ in G (and in A), because both $\bar{x}_{I'}$ and x are above C^G_{suf}.

The equalities

$$(\lambda,I')\text{-dist}(x) = (\lambda,I')\text{-dist}(\bar{x}_{I'}) = (\lambda,I')\text{-dist}(x^{\mathrm{close}}_{I'}) = 0$$

are established analogously to Lemma 3.14, by considering the distance of $\bar{x}_{I'}$ to the λ-case I' and a segmented virtual execution $\bar{x}_{I'} \overset{*}{\dashrightarrow} x$ in $A' = A^G_{\mathrm{SC}}$.

For each $i \in I'$, the values $x(i)$ and $\bar{x}_{I'}(i) = x^{\mathrm{close}}_{I'}(i)$ are in the interval $[C^G_{\mathrm{suf}}, \lambda_{|I'|}-1]$; therefore, $||x_{|I'} - (\bar{x}_{I'})_{|I'}|| \leq \lambda_{|I'|} - C^G_{\mathrm{suf}}$. The fact $(p_0, \bar{x}_{I'}) \xrightarrow{*} (p_0, x)$ implies that there exists an action sequence σ_2 such that $(p_0, (\bar{x}_{I'})_{|I'}) \xrightarrow{\sigma_2} (p_0, x_{|I'})$ in the restricted PNS $G_{|I\cup I'}$ (that is, in $A_{|I\cup I'}$) and

$$|\sigma_2| \leq f_{\mathrm{vr}}(||G||, |I \cup I'|) \cdot ||x_{|I'} - (\bar{x}_{I'})_{|I'}|| \leq C^G_{\mathrm{vr}} \cdot (\lambda_{|I'|} - C^G_{\mathrm{suf}}).$$

Combining with $(p_0, x) \xrightarrow{\sigma_1} (q, y)$, where $|\sigma_1| \leq C^G_{\mathrm{qd}}$ and (q, y) is dead, we obtain

$$(p_0, (\bar{x}_{I'})_{|I'}) = (p_0, (x^{\mathrm{close}}_{I'})_{|I'}) \xrightarrow{\sigma_2} (p_0, x_{|I'}) \xrightarrow{\sigma_1} (q, y_{|I'})$$

in $G_{|I\cup I'}$, where $|\sigma_2\sigma_1| \leq C^G_{\mathrm{vr}} \cdot (\lambda_{|I'|} - C^G_{\mathrm{suf}}) + C^G_{\mathrm{qd}}$. Since for each $i \in J \smallsetminus I'$, we have $x^{\mathrm{close}}_{I'}(i) \geq \lambda_{|I'|+1} \geq ||A|| \cdot |\sigma_2\sigma_1|$, it follows that

$$(p_0, x^{\mathrm{close}}_{I'}) \xrightarrow{\sigma_2\sigma_1} (q, y')$$

is an execution in G, where $y'_{|I'} = y_{|I'}$. We complete the proof by showing that (q, y') is dead, which contradicts the assumption that $(p_0, x^{\mathrm{close}}_{I'})$ is live (since $(p_0, x_0) \xrightarrow{*} (p_0, x^{\mathrm{close}}_{I'})$ and (p_0, x_0) is live).

Recall that $(p_0, x) \xrightarrow{\sigma_1} (q, y)$ and $|\sigma_1| \leq C^G_{\mathrm{qd}}$. Since for each $i \in J \smallsetminus I'$, we have $x(i) \geq \lambda_{|I'|+1} \geq \lambda_1 \geq C^A_{\mathrm{dead}} + ||A|| \cdot C^G_{\mathrm{qd}}$, it follows that $y(i) \geq C^A_{\mathrm{dead}}$ for each $i \in J \smallsetminus I'$. This implies that $(q, y_{|I'})$ is dead in $A_{|I\cup I'}$, which in turn implies that (q, y') is dead in A (since $y'_{|I'} = y_{|I'}$). □

Proof of Theorem 2.13. Given a d-dim net A possessing a live bottom SCC $X \subseteq \mathbb{N}^d$, we consider a feasible PNS $G = G^{(A,\lambda')}_{(I,Q)}$ as defined in Definition 3.32. We set $J = [1, d] \smallsetminus I$.

Since $||Q|| < \lambda'_{|I|}$, it follows that

$$||G|| = \max\{||Q||, ||A||\} \leq (2 + ||A||)^{2^{\mathrm{poly}(d)}}.$$

The 2-exp bound follows from the definition of λ' and the closure properties of 1-exp and 2-exp functions discussed at the beginning of this section. Moreover, by Proposition 3.33, there exists a live marking $(p_0, x_0) \in X$ with $p_0 \in Q = X_{|I}$ and

$$x_0 \in \uparrow \left(2 \cdot (|Q| - 1) \cdot ||A|| + \max\{C^G_{\mathrm{suf}}, ||A|| \cdot C^G_{\mathrm{qd}}\}\right).$$

Recall that $|Q| \leq (1 + ||Q||)^{|I|}$ (where $|I| \leq d$) and that f_{suf} and f_{qd} are 2-exp functions. Lemma 3.34 implies that every solution of a $(G, (p_0, x_0), \lambda)$-linear system S, where λ is defined by

$$\lambda_1 = C^G_{\mathrm{suf}} + ||A|| \cdot C^G_{\mathrm{qd}}, \text{ and } \lambda_{i+1} = \lambda_1 + ||A|| \cdot C^G_{\mathrm{vr}} \cdot (\lambda_i - C^G_{\mathrm{suf}}) \text{ for each } i \in [1, |J|-1],$$

yields a live marking $(p_0, \bar{x}_0)$. The dimension of S is at most $d \cdot 2^d$, and

$$\max\{\mathrm{lcm}(S), ||S||\} \leq (2 + ||A||)^{2^{\mathrm{poly}(d)}}.$$

Indeed, the virtual reachability in S refers to the "simple-cycle net" A^G_{SC}; consequently, Corollary 2.12 implies that

$$\mathrm{lcm}(S) \leq |J|\,! \cdot ||A^G_{\mathrm{SC}}|| \leq d\,! \cdot |Q| \cdot ||A|| \leq d\,! \cdot (1 + ||Q||)^d \cdot ||A||,$$

where $||Q|| \leq (2+||A||)^{2^{\text{poly}(d)}}$. Since $B = B_{(G,\lambda)} = \max\{C^G_{\text{suf}}, \lambda_{|J|} + C^{A^G_{\text{SC}}}_{\text{shift}}\}$, where

$$\lambda_{|J|} \leq (2+||A||)^{2^{\text{poly}(|J|)}} \leq (2+||A||)^{2^{\text{poly}(d)}} \text{ and } f_{\text{shift}}(||A^G_{\text{SC}}||, |J|) \leq f_{\text{shift}}(|Q| \cdot ||A||, d),$$

we also obtain $||S|| \leq (2+||A||)^{2^{\text{poly}(d)}}$.

Analogously to Proposition 3.9, we use Theorem 2.9 to deduce that there exists a solution of S that yields $(p_0, \bar{x}_0)$ (which is a live marking of A) such that

$$||(p_0, \bar{x}_0)|| \leq (2+||A||)^{2^{\text{poly}(d)}}. \qquad \square$$

3.2. **Dead markings.**

In this section, we provide further details regarding the characterisation (3) of dead markings. Recall that in our terminology, a marking $y \in \mathbb{N}^d$ of a d-dim net A is dead if there exists an action $a \in A$ that is dead at y, that is, a is disabled at every marking $y' \in \text{Reach}(y)$. The following proposition is a straightforward observation.

Proposition 3.35 (**If a restriction of a marking is dead, then the marking is dead**). *Given a d-dim net A and $I \subseteq [1,d]$, if $y_{|I}$ is dead in $A_{|I}$, then y is dead in A.*

Proof. The claim follows from the fact that, for every $y \in \mathbb{N}^d$, the set $\{y'_{|I} \mid y \xrightarrow{*} y' \text{ in } A\}$ is a subset of the set $\{y' \in \mathbb{N}^I \mid y_{|I} \xrightarrow{*} y' \text{ in } A_{|I}\}$. $\square$

We now adapt Rackoff's result for coverability from [Rac78].

Lemma 3.36 (**Deadness is determined by components with "small" values**).
There exists a 2-exp function f_{dead} with the following property:
For every d-dim net A, a marking $y \in \mathbb{N}^d$ is dead in A if and only if $y_{|\text{s}_y}$ is dead in the restricted net $A_{|\text{s}_y}$, where $\text{s}_y = \{i \in [1,d] \mid y(i) < f_{\text{dead}}(||A||, d)\}$.

Proof. Note that an action $a = (a_-, a_+)$ in a net A is not dead at a marking x if and only if x can cover a_-, that is, if and only if there exists x' such that $x \xrightarrow{*} x' \geq a_-$. Along the lines of [Rac78] (Lemmas 3.3. and 3.4), we define a function $g : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ as follows:

- $g(m,0) = 0$;
- for $i \geq 1$, $g(m,i) = (m + m \cdot g(m, i-1))^i + g(m, i-1)$.

By induction on $|I|$, it is straightforward to verify that for a d-dim net A, an action $a = (a_-, a_+) \in A$, an index set $I \subseteq [1,d]$, and a marking $x \in \mathbb{N}^I$ of $A_{|I}$, the following holds: if x can cover $(a_-)_{|I}$ (in the net $A_{|I}$), then there exist an action sequence σ and a marking $x' \in \mathbb{N}^I$ such that

$$x \xrightarrow{\sigma} x' \geq (a_-)_{|I} \text{ and } |\sigma| \leq g(||A||, |I|).$$

The claim is trivial for $I = \emptyset$. In the case $I \neq \emptyset$, we observe that if a shortest execution $x \xrightarrow{*} x' \geq (a_-)_{|I}$ in $A_{|I}$ has a prefix reaching the value $n = ||A|| + ||A|| \cdot g(||A||, |I|-1)$ or larger in some component, then the length of the suffix is bounded by $g(||A||, |I|-1)$ by the inductive hypothesis, and the length of the prefix is bounded by $n^{|I|}$.

This implies that for any d-dim net A and $x \in \mathbb{N}^d$, the values $x(i)$, $i \in [1,d]$, for which $x(i) \geq ||A|| + ||A|| \cdot g(||A||, d)$ are irrelevant to the question of whether x is dead. As the required 2-exp function, we can thus take

$$f_{\text{dead}}(m,i) = m + m \cdot g(m,i). \qquad \square$$

Remark 3.37. We note that the recent paper [KMS+23] has improved Rackoff's bound in [Rac78], but such fine-tuned results are not needed at our level of analysis.

3.3. **Virtual reachability in reversible nets.**

In this section, we formalize the previous informal claim regarding small-segment virtual executions (5) that are "directed" from a marking x to a marking y.

We recall the notation $A_\delta = \{\Delta(a) \mid a \in A\}$ and the fact that the virtual reachability relation $\overset{*}{\dashrightarrow}$ in any *reversible* d-dim net A is symmetric: $x \overset{*}{\dashrightarrow} y$ implies $y \overset{*}{\dashrightarrow} x$. This follows from the fact that $x \overset{*}{\dashrightarrow} y$ in A is equivalent to $(y-x) \in A_\delta^*$, and that A_δ^* is a subgroup of $(\mathbb{Z}^d, +)$ if A is reversible.

For any $d \in \mathbb{N}$, we define the function

$$\textsc{sign} : \mathbb{Z}^d \to \{-1,0,1\}^d$$

as follows: for $x \in \mathbb{Z}^d$ and $i \in [1,d]$, we set

- $\textsc{sign}(x)(i) = -1$ if $x(i) < 0$;
- $\textsc{sign}(x)(i) = 0$ if $x(i) = 0$;
- $\textsc{sign}(x)(i) = 1$ if $x(i) > 0$.

We define the following partial order $\preceq$ on $\mathbb{Z}^d$:

$$x \preceq y \text{ if } \textsc{sign}(x) = \textsc{sign}(y) \text{ and } |x(i)| \leq |y(i)| \text{ for all } i \in [1,d].$$

For instance, $(0,5,-3) \preceq (0,11,-6)$, but $(0,5,-3) \not\preceq (1,11,-6)$, since $\textsc{sign}((0,5,-3)) = (0,1,-1) \neq \textsc{sign}((1,11,-6)) = (1,1,-1)$. Note that $\mathbf{0} \preceq x$ implies $x = \mathbf{0}$. For $X \subseteq \mathbb{Z}^d$,

$$\min_{\preceq}(X)$$

denotes the set of minimal elements of X w.r.t. $\preceq$; it is finite since $\preceq$ is a well-partial-order.

Proposition 3.38 ("Directed" decomposition of elements in group L by $\min_{\preceq}(L)$). *If $L \subseteq \mathbb{Z}^d$ is a group and $y \in L$, then there exists $z \in \min_{\preceq}(L)$ such that $z \preceq y$, $y = z + (y-z)$, and $(y-z) \in L$. Moreover, $||y-z|| < ||y||$ if $y \neq \mathbf{0}$.*

Proof. Let $L \subseteq \mathbb{Z}^d$ be a group, and let $y \in L$. Since $\preceq$ is a well-partial-order, there exists $z \in \min_{\preceq}(L)$ such that $z \preceq y$. We have $(y-z) \in L$ because L is a group. To show the strict decrease in norm, consider a component $i \in [1,d]$:

- If $y(i) = 0$, then $z(i) = 0$, since $\textsc{sign}(y) = \textsc{sign}(z)$; therefore, $y(i) - z(i) = 0$.
- If $y(i) > 0$, then $0 < z(i) \leq y(i)$, which implies $0 \leq y(i) - z(i) < y(i)$; therefore, $|y(i) - z(i)| < |y(i)|$.
- If $y(i) < 0$, then $y(i) \leq z(i) < 0$, which implies $y(i) < y(i) - z(i) \leq 0$; therefore, $|y(i) - z(i)| < |y(i)|$.

Consequently, $||y-z|| < ||y||$ if $y \neq \mathbf{0}$. $\square$

Corollary 3.39 (**Segmenting virtual executions of reversible nets**).
Given a reversible net A (for which A^*_δ is a group), if $x \xrightarrow{*} y$, then there exist $z_1, z_2, \ldots, z_k$ in $\min_\preceq(A^*_\delta)$ such that

$$x \xrightarrow{*} (x + z_1) \xrightarrow{*} (x + z_1 + z_2) \cdots \xrightarrow{*} (x + z_1 + z_2 \cdots + z_k) = y \tag{34}$$

and for each $j \in [1, k]$ it holds that $z_j \preceq y - (x + z_1 + z_2 \cdots + z_{j-1})$.

Proof. This is a direct consequence of Proposition 3.38, since $x \xrightarrow{*} y$ is equivalent to $(y - x) \in A^*_\delta$. By applying the proposition repeatedly, we construct the sequence: in the first step, we choose $z_1 \in \min_\preceq(A^*_\delta)$ such that $z_1 \preceq (y-x)$. Thus $y-x = z_1+(y-x-z_1)$, which implies $y-x-z_1 = y-(x+z_1) \in A^*_\delta$; that is, $(x + z_1) \xrightarrow{*} y$. Moreover, if $x \neq y$, then $||y - (x + z_1)|| < ||y - x||$. □

Recall that we used $C^A_{\text{shift}} = f_{\text{shift}}(||A||, d)$ to bound the norms of segments in the virtual executions (5) of a d-dim net A. In principle, by (6) we claimed that $||\min_\preceq(A^*_\delta)|| \leq f_{\text{shift}}(||A||, d)$, where f_{shift} is a 1-exp function. The following lemma establishes this result; it is slightly more general, since it shows that

$$||\min_\preceq(A^*)|| \leq (2 + ||A||)^{\text{poly}(d)}$$

for every finite set $A \subseteq \mathbb{Z}^d$ (where A^* is not necessarily a group).

Lemma 3.40 (**For every finite set $A \subseteq \mathbb{Z}^d$, the norm $||\min_\preceq(A^*)||$ is 1-exp**).
For every finite set $A \subseteq \mathbb{Z}^d$, it holds that

$$\|\min_\preceq(A^*)\| \leq (3 + ||A|| \cdot |A|)^d \leq (3 + ||A|| \cdot (1 + 2 \cdot ||A||)^d)^d.$$

Proof. By decomposing the set A^* according to the function SIGN (recall that $x \preceq y$ requires $\text{SIGN}(x) = \text{SIGN}(y)$, so vectors in different sign classes are incomparable), we obtain:

$$\min_\preceq(A^*) = \bigcup_{s \in \{-1,0,1\}^d} \min_\preceq(\{y \in A^* \mid \text{SIGN}(y) = s\}).$$

Thus, it suffices to provide the required bound on $||\min_\preceq(\{y \in A^* \mid \text{SIGN}(y) = s\})||$ for an arbitrary vector $s \in \{-1, 0, 1\}^d$, which is henceforth fixed. Observe that by flipping the sign of the ith component of each vector in A for all $i \in [1, d]$ such that $s(i) = -1$, we may assume without loss of generality that $s \in \{0, 1\}^d$. We set

$$I = \{i \in [1, d] \mid s(i) = 0\}.$$

For $k = |A|$, we henceforth view A as a $d \times k$ matrix (with d rows and k columns), where the columns correspond to the elements of A. Consequently, we now aim to provide a bound on the norm $||\min_\leq(Y)||$, where $\leq$ denotes the standard component-wise order and the set $Y \subseteq \mathbb{N}^d$ is defined as follows:

$$Y = \{Av \mid v \in \mathbb{N}^k \wedge \bigwedge_{i \in I} (Av)(i) = 0 \wedge \bigwedge_{i \in [1,d] \setminus I} (Av)(i) \geq 1\}.$$

We proceed similarly to the proof of [Pot91, Corollary 1], which deals with systems of linear inequalities. Let B be the $d \times (k + d + 1)$ matrix

$$B = \left(\begin{array}{c|c|c} A & -\text{Id}_d & -s \end{array} \right),$$

that is, the k columns of A, followed by the columns of the negated $d \times d$ identity matrix, followed by the single column $-s$. We set

$$M = \{x \in \mathbb{N}^{k+d+1} \mid Bx = \mathbf{0}\} = \{(v, z, t) \in \mathbb{N}^k \times \mathbb{N}^d \times \mathbb{N} \mid Av - z - ts = \mathbf{0}\},$$

where the three blocks of B act on v, z, and t, respectively. We then observe that

$$Y = \{Av \mid \exists z \in \mathbb{N}^d : (v, z, 1) \in M \wedge \bigwedge_{i \in I} z(i) = 0\}. \tag{35}$$

Lemma 2.8 implies that $M = X^*$, where $X = \min_{\leq}(M \setminus \{\mathbf{0}\})$ and, since $\mathrm{rank}(B) \leq d$ and $\|B\|_1 \leq k \cdot \|A\| + 2$,

$$\|X\|_1 \leq (3 + k \cdot \|A\|)^d. \tag{36}$$

Finally, we fix an arbitrary $y \in \min_{\leq}(Y)$ and bound its norm $||y||$. By (35), we may fix v and z such that $y = Av$, $(v, z, 1) \in M$, and $z(i) = 0$ for all $i \in I$. Since $M = X^*$, the tuple $(v, z, 1)$ is a sum of elements of X; their last components are nonnegative and sum to 1, so exactly one summand $(v', z', 1)$ has last component 1, and it satisfies $(v', z') \leq (v, z)$ and $\|(v', z', 1)\|_1 \leq (3 + k \cdot \|A\|)^d$ by (36). As $z' \leq z$ and $z(i) = 0$ for all $i \in I$, we have $z'(i) = 0$ for all $i \in I$, and hence $Av' = s + z'$ belongs to Y by (35). Because

$$Av' = s + z' \leq s + z = Av = y$$

and $y \in \min_{\leq}(Y)$, it follows that $z' = z$. Consequently, discarding the last component of $(v', z', 1)$,

$$||z|| \leq \|z\|_1 = \|z'\|_1 \leq (3 + k \cdot \|A\|)^d - 1.$$

Since $Av - z = y - z = s \in \{0, 1\}^d$, we conclude that $||y|| \leq ||z|| + 1 \leq (3 + k \cdot \|A\|)^d$. □

4. Bounds on Minimal Solutions of Linear Systems

In this section, we prove Theorem 2.9, which we repeat here for convenience.

Theorem 2.9 (**Bounds on solutions of linear systems**).
Every satisfiable d-dim linear system S has a solution $x \in \mathbb{Z}^d$ such that

$$||x|| \leq \mathrm{lcm}(S) \cdot \left(1 + (d+1)! \cdot (d \cdot ||S|| + 1)^d\right).$$

We first prove the theorem for the case where the linear system S does not contain any divisibility constraints. We use the following result, which is a weaker form of the theorem from [vzGS78].

Lemma 4.1 [vzGS78] (**Bounds on solutions to inequalities via subdeterminants**). *Let C and c be $n \times d$ and $n \times 1$ integer matrices, respectively, and let r be the rank of C. Let m be an upper bound on the absolute values of the $(r-1) \times (r-1)$ and $r \times r$ subdeterminants of the $n \times (d+1)$ augmented matrix (C, c). If $Cx \geq c$ has an integer solution, then it has a solution $x \in \mathbb{Z}^d$ such that $|x(i)| \leq (d+1) \cdot m$ for all $i \in [1, d]$.*

Lemma 4.2 (**Systems without divisibility constraints**).
Every satisfiable d-dim linear system S without divisibility constraints has a solution $x \in \mathbb{Z}^d$ such that

$$||x|| \leq (d+1)! \cdot (||S|| + 1)^d.$$

Proof. Let S be a satisfiable d-dim linear system, expressed as a propositional formula whose atoms are equality and inequality constraints (with no divisibility constraints). Each equality constraint $(\alpha \cdot \mathbf{x}) = b$ in S can be replaced by the conjunction $(\alpha \cdot \mathbf{x}) \geq b \wedge (-\alpha \cdot \mathbf{x}) \geq -b$ without affecting $[\![S]\!]$ or $||S||$.

We may thus assume, without loss of generality, that the atoms in S consist only of inequality constraints. By expressing S in disjunctive normal form, we obtain linear systems $S_1, S_2, \ldots, S_k$, each given as a conjunction of inequalities and negations of inequalities, where

$$[\![S]\!] = \bigcup_{j=1}^{k} [\![S_j]\!], \text{ and } ||S_j|| \leq ||S|| \text{ for all } j \in [1,k].$$

Since $[\![S]\!] \neq \emptyset$, it follows that $[\![S_j]\!] \neq \emptyset$ for some $j \in [1,k]$. It therefore suffices to establish the claim for the case where S is a conjunction of inequality constraints and their negations; we will assume this henceforth.

Let S' be the linear system obtained from S by replacing each subformula $\neg((\alpha \cdot \mathbf{x}) \geq b)$ with the inequality constraint $(-\alpha \cdot \mathbf{x}) \geq -b+1$. Then

$$[\![S']\!] = [\![S]\!] = \{x \in \mathbb{Z}^d \mid Cx \geq c\}, \tag{37}$$

where C and c are some $n \times d$ and $n \times 1$ integer matrices, respectively, with

$$\max\{||C||, ||c||\} = ||S'|| \leq ||S|| + 1.$$

Let r be the rank of C, and let m be an upper bound on the absolute values of the $(r-1)\times(r-1)$ and $r \times r$ subdeterminants of the $n \times (d+1)$ augmented matrix (C, c). Since $r \leq d$, it follows that

$$m \leq d\,! \cdot (||S|| + 1)^d.$$

Applying Lemma 4.1 to (37), we conclude that S' admits a solution x such that

$$||x|| \leq (d+1) \cdot m \leq (d+1)\,! \cdot (||S|| + 1)^d. \qquad \square$$

Proof of Theorem 2.9. Let S be a satisfiable d-dim linear system, with a variable vector $\mathbf{x}$ ranging over $\mathbb{Z}^d$. We set $\ell = \mathrm{lcm}(S)$ and fix a solution SOL of S. Let

$$b \in \{0, 1, \ldots, \ell-1\}^d \text{ be the vector of remainders of SOL modulo } \ell;$$

specifically, we define $b(i) = (\mathrm{SOL}(i) \bmod \ell)$ for all $i \in [1,d]$.

We aim to construct a d-dim linear system S_b, with a variable vector $\mathbf{y}$ and without divisibility constraints, such that

$$[\![S_b]\!] = \{y \in \mathbb{Z}^d \mid b + \ell y \in [\![S]\!]\}. \tag{38}$$

We construct S_b from S by replacing each constraint as follows:

- $(\alpha \cdot \mathbf{x}) = c$ is replaced by $(\alpha \cdot \mathbf{y}) = \frac{c-(\alpha\cdot b)}{\ell}$ if ℓ divides $c - (\alpha \cdot b)$, and by *false* otherwise;
- $(\alpha \cdot \mathbf{x}) \geq c$ is replaced by $(\alpha \cdot \mathbf{y}) \geq \left\lceil \frac{c-(\alpha\cdot b)}{\ell} \right\rceil$;
- $(\alpha \cdot \mathbf{x}) \equiv c \,(\mathrm{mod}\, m)$ is replaced by *true* if $(\alpha \cdot b) \equiv c \,(\mathrm{mod}\, m)$ and by *false* otherwise.

It is straightforward to verify that $[\![S_b]\!]$ satisfies the condition (38), by noting that $y \in [\![\text{new constraint}]\!]$ if and only if $x \in [\![\text{old constraint}]\!]$ for $x = b + \ell y$ (using the fact that ℓ is a multiple of each m appearing in the divisibility constraints). Moreover, $[\![S_b]\!] \neq \emptyset$ because the vector y defined by $y(i) = \frac{\text{SOL}(i) - b(i)}{\ell}$ for each $i \in [1, d]$ is a solution.

Note that if $(\alpha \cdot \mathbf{x}) \sim c$ is a constraint of S, where $\sim$ is the equality or the inequality, then the values $|\alpha(i)|$ for $i \in [1, d]$ and $|c|$ are bounded by $||S||$. It follows that $|(\alpha \cdot b)| \leq d \cdot ||S|| \cdot (\ell - 1)$, and $|\frac{c - (\alpha \cdot b)}{\ell}| \leq d \cdot ||S||$. We have thus shown that

$$||S_b|| \leq d \cdot ||S||.$$

Since S_b is satisfiable and contains no divisibility constraints, Lemma 4.2 implies that S_b has a solution $y \in \mathbb{Z}^d$ such that

$$||y|| \leq (d+1)\,! \cdot (||S_b|| + 1)^d.$$

Hence $x = b + \ell y$ is a solution of S, satisfying

$$||x|| \leq (\ell - 1) + \ell \cdot (d+1)\,! \cdot (||S_b|| + 1)^d \leq \ell \cdot \left(1 + (d+1)\,! \cdot (d \cdot ||S|| + 1)^d\right).$$

□

5. Bounds on Sizes of Linear Systems for Groups

In this section, we prove Theorem 2.11, which we repeat here for convenience.

Theorem 2.11 (**Groups expressed by linear systems of bounded size**).
Let L be the group spanned by a finite set $X \subseteq \mathbb{Z}^d$. There exists a d-dim linear system S such that

$$[\![S]\!] = L \text{ and } \max\{||S||, \text{lcm}(S)\} \leq d\,! \cdot \|X\|^d.$$

Recall that the group L spanned by a finite set $X \subseteq \mathbb{Z}^d$ is the set of integer linear combinations of elements of X. The elements of X can be viewed as the columns of an integer $d \times k$ matrix C, where $k = |X|$; in this case, $||X|| = ||C||$.

To prove the theorem, we consider an integer $d \times k$ matrix C and denote by L_C the group (also called a *lattice*) spanned by the columns of C; that is,

$$L_C = \{x \in \mathbb{Z}^d \mid \exists y \in \mathbb{Z}^k : Cy = x\}.$$

We aim to construct a linear system S such that $[\![S]\!] = L_C$ and

$$\max\{||S||, \text{lcm}(S)\} \leq d\,! \cdot ||C||^d. \tag{39}$$

Case $\text{rank}(\boldsymbol{C}) = \boldsymbol{d}$. We first consider the case where $\text{rank}(C) = d$; that is, we assume that the rows of C are linearly independent. From [Sch86, Chapter 4], we recall that there exists a square $d \times d$ non-singular integer matrix H such that $L_H = L_C$. We can take H to be the first d columns of the Hermite normal form of C (the remaining $k - d$ columns being zero vectors). It follows that

$$L_C = L_H = \{x \in \mathbb{Z}^d \mid \exists y \in \mathbb{Z}^d : Hy = x\} = \{x \in \mathbb{Z}^d \mid H^{-1}x \in \mathbb{Z}^d\};$$

note that $H^{-1}x \in \mathbb{Z}^d$ implies that $x = H(H^{-1}x) \in \mathbb{Z}^d$.

Although the entries of H^{-1} may be non-integers, the matrix $\det(H) \cdot H^{-1}$ is an integer matrix, since $H^{-1} = \frac{1}{\det(H)} \cdot \text{adj}(H)$, where $\text{adj}(H)$ is the adjugate of H (and thus an integer matrix). In particular, $H^{-1}(\det(H) \cdot z)$ is an integer vector for any $z \in \mathbb{Z}^d$. This implies that for any $x \in \mathbb{Z}^d$,

$$\text{the vector } H^{-1}x \text{ is an integer vector if and only if } H^{-1}(x \bmod \ell) \text{ is an integer vector,}$$

where $\ell = |\det(H)|$ and the modulo operation is applied component-wise.

Defining $B = \{b \in \{0, 1, \ldots, \ell-1\}^d \mid H^{-1}b \in \mathbb{Z}^d\}$, we obtain $L_C = [\![S]\!]$, where

$$S = \bigvee_{b \in B} \left(\bigwedge_{i=1}^{d} \mathbf{x}_i \equiv b(i) \pmod{\ell} \right);$$

here $\mathbf{x} = (\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_d)$ is the variable vector of S ranging over $\mathbb{Z}^d$.

To ensure the bound (39), it suffices to show that $|\det(H)| \leq d! \cdot ||C||^d$. Consider any $d \times d$ matrix M composed of d linearly independent columns of C; hence L_M is a subgroup of $L_C = L_H$. From [Sch86, Chapter 4], it follows that $\det(H)$ divides $\det(M)$, which implies that $|\det(H)| \leq |\det(M)|$. Since $|\det(M)| \leq d! \cdot ||M||^d$ and $||M|| \leq ||C||$, we have completed the case for $\operatorname{rank}(C) = d$.

Case $\operatorname{rank}(\boldsymbol{C}) = \boldsymbol{r} < \boldsymbol{d}$. We now consider a $d \times k$ integer matrix C with $\operatorname{rank}(C) = r < d$. Without loss of generality, we assume (possibly after reordering the rows of C) that C is of the form

$$C = \begin{bmatrix} \bar{C} \\ C' \end{bmatrix},$$

where $\bar{C}$ is an $r \times k$ matrix with $\operatorname{rank}(\bar{C}) = r$ (that is, $\bar{C}$ is a row basis of C), and each of the $d - r$ rows of C' is a rational linear combination of the r rows of $\bar{C}$. Hence, there exists a unique rational $(d - r) \times r$ matrix T such that $C' = T\bar{C}$.

We now examine the transformation of $\bar{C}$ into its Hermite normal form in more detail. This is achieved by a sequence of elementary column operations on $\bar{C}$, comprising: (i) multiplying a column by -1, (ii) swapping two columns, and (iii) adding an integer multiple of one column to another. These operations correspond to multiplying $\bar{C}$ from the right by a unimodular matrix U (an integer $k \times k$ matrix with $|\det(U)| = 1$). Multiplying C by U, we obtain

$$CU = \begin{bmatrix} \bar{C} \\ C' \end{bmatrix} U = \begin{bmatrix} \bar{H} & 0 \\ H' & 0 \end{bmatrix} = [H \mid 0],$$

where $\bar{H}$ is a non-singular $r \times r$ integer matrix such that $L_{\bar{H}} = L_{\bar{C}}$, and $H' = T\bar{H}$. The latter holds because $C' = T\bar{C}$ and elementary column operations preserve linear relations between rows.

Since elementary column operations preserve the spanned group, we have

$$L_C = L_H = \{x \in \mathbb{Z}^d \mid \exists y \in \mathbb{Z}^r : \bar{H}y = x_{|[1,r]} \wedge H'y = x_{|[r+1,d]}\}.$$

By rewriting the conjunction as $y = \bar{H}^{-1}x_{|[1,r]} \wedge H'\bar{H}^{-1}x_{|[1,r]} = x_{|[r+1,d]}$, and noting that $H'\bar{H}^{-1} = T$, we obtain

$$L_C = \{x \in \mathbb{Z}^d \mid x_{|[1,r]} \in L_{\bar{C}} \wedge x_{|[r+1,d]} = Tx_{|[1,r]}\}. \tag{40}$$

We note that the rational $(d-r) \times r$ matrix T is determined by any submatrix M consisting of r linearly independent columns of C. Letting $\bar{M} = M_{|[1,r]}$ and $M' = M_{|[r+1,d]}$, we have $M' = T\bar{M}$ (as $C' = T\bar{C}$), which implies that the r rows of $\bar{M}$ are linearly independent; hence, $\bar{M}$ is non-singular. Using Cramer's rule, we deduce that all entries of T are of the form $\frac{\det(M'')}{\det(\bar{M})}$, with the same denominator $\det(\bar{M})$ and varying integer numerators. The absolute values of the denominator and all numerators are bounded by $r! \cdot ||M||^r$, where $r < d$ and $||M|| \leq ||C||$.

The condition $x_{|[r+1,d]} = Tx_{|[1,r]}$ can be expressed as a conjunction of homogeneous linear constraints for $i \in [r+1, d]$ of the form

$$\det(\bar{M}) \cdot \mathbf{x}_i - \sum_{j=1}^{r} \det(M''_{ij}) \cdot \mathbf{x}_j = 0,$$

where the absolute values of all coefficients are bounded by $r! \cdot ||C||^r$. Since $r < d$, the group L_C in (40) can indeed be expressed by a linear system S satisfying (39). Here, we apply the results from the case $\mathrm{rank}(C) = d$ to the submatrix $\bar{C}$ with $\mathrm{rank}(\bar{C}) = r$ (noting that $||\bar{C}|| \leq ||C||$). This completes the proof of Theorem 2.11.

6. EXPSPACE-hardness (a counterpart of Theorem 2.13)

In this section, we prove the following theorem for a subclass of 1-conservative nets. A net A is a *population protocol net* (a *PP-net* for short) if each action (a_-, a_+) in A satisfies either $||a_-||_1 = ||a_+||_1 = 1$ or $||a_-||_1 = ||a_+||_1 = 2$. A d-dim *net* A is *ordinary* if $A \subseteq \{0,1\}^d \times \{0,1\}^d$.

Theorem 6.1 (**EXPSPACE-hardness for a simple subclass of 1-conservative nets**). *The structural liveness problem is EXPSPACE-hard for ordinary reversible PP-nets.*

Remark 6.2. It is evident that PP-nets are indeed a subclass of 1-conservative nets; they model the *population protocols* introduced in [AAD$^+$06]. In these protocols, an arbitrary number of pairwise indistinguishable finite-state agents interact in pairs. Here, a marking is a function that assigns to each state q the number of agents currently in that state. When two agents in states q_1 and q_2 interact, they change their states to q_3 and q_4, according to a transition function; the states q_1, q_2, q_3, q_4 are not necessarily pairwise distinct. In [JV23], the structural liveness was studied for *immediate observation nets*. This is a subclass of PP-nets modelling *immediate observation protocols* introduced in [AAER07], and for their generalised variants; these classes of nets were introduced and studied in [ERW19, RW20, RWE20, Wei23]. The structural liveness problem for these classes was shown to be PSPACE-complete [JV23]; Theorem 6.1 thus demonstrates that the problem for the class of PP-nets is substantially harder, even when restricted to ordinary and (structurally) reversible PP-nets.

A useful EXPSPACE-hard problem Cover. Our proof of Theorem 6.1 draws on ideas from [JP19]. First, we recall the well-known result presented in [MM82] showing that the uniform word problem for commutative semigroups is EXPSPACE-complete. An instance of this problem consists of a finite alphabet Σ, a finite set of equations $u \equiv v$ for $u, v \in \Sigma^*$ (implicitly comprising $ab \equiv ba$ for all $a, b \in \Sigma$), and $u_0, v_0 \in \Sigma^*$; the question is whether $u_0 \equiv v_0$. The crux of the high complexity lies in the fact that the commutative semigroup defined by $a \equiv b^{2^{2^n}}$, which can be written in space $O(2^n)$ when using binary notation for exponents, can be embedded in a commutative semigroup of size $O(n)$, even when using unary notation for exponents. In fact, even the (weaker) *coverability* version in which $u_0, v_0 \in \Sigma$ and we ask whether $u_0 \equiv v_0 w$ for some $w \in \Sigma^*$ is EXPSPACE-complete. Moreover, we can assume that in each given equation $u \equiv v$ neither u nor v contains more than one occurrence of any $a \in \Sigma$. (The crux is that $aau' \equiv v$ can be replaced with $a_1 a_2 u' \equiv v$, and $a_1 \equiv a$, $a_2 \equiv a$.)

It is straightforward to formulate the aforementioned EXPSPACE-complete coverability problem in terms of Petri nets. We say that a *net* A is *strongly reversible* if $A = A^{-1}$ (hence if $(a_-, a_+) \in A$ implies $(a_+, a_-) \in A$). Given $d \in \mathbb{N}$, for each $i \in [1, d]$, we denote by $\mathbf{e}_i$ the unit vector from $\{0,1\}^d$ satisfying $\mathbf{e}_i(i) = 1$ and $\mathbf{e}_i(j) = 0$ for all $j \neq i$. Recall that, in our context, the term *places* is a synonym for *vector components*.

Proposition 6.3 (**Problem** Cover **for ordinary strongly reversible nets**).
The following problem Cover *is EXPSPACE-hard:*

Instance: An ordinary strongly reversible d-dim net A, and two places INIT, COV $\in [1,d]$.
Question: Does there exist $y \in \mathbb{N}^d$ *such that* $\mathbf{e}_{\text{INIT}} \xrightarrow{*} (\mathbf{e}_{\text{COV}} + y)$ *?*

Proof. An instance of the aforementioned coverability problem for commutative semigroups, with equations $u \equiv v$ over Σ and distinguished elements $u_0, v_0 \in \Sigma$, can be viewed as an instance of COVER: We consider a bijection B from Σ onto $\{1, 2, \ldots, |\Sigma|\}$, and the $|\Sigma|$-dim net where each equation $u \equiv v$ gives rise to two actions (χ_u, χ_v) and (χ_v, χ_u); for the characteristic vector $\chi_w \in \{0,1\}^{|\Sigma|}$ we have $\chi_w(i) = 1$ iff $\text{B}^{-1}(i)$ occurs in w. Finally, we set INIT $= \text{B}(u_0)$ and COV $= \text{B}(v_0)$. □

Reducing COVER to SCOVER for **1**-conservative nets. We prove Theorem 6.1 by showing a polynomial reduction from the problem COVER to the structural liveness problem for ordinary reversible PP-nets. For convenience, we use two intermediate problems, called SCOVER and PPSCOVER. The letter S refers to *store places*, which enable a simple transformation of COVER to a corresponding problem for 1-conservative nets.

PROBLEM SCOVER

***Instance*:** An ordinary strongly reversible 1-conservative d-dim net A, and three places INIT, COV, STORE $\in [1,d]$.
***Question*:** Do there exist $k \in \mathbb{N}$ and $y \in \mathbb{N}^d$ such that

$$(\mathbf{e}_{\text{INIT}} + k \cdot \mathbf{e}_{\text{STORE}}) \xrightarrow{*} (\mathbf{e}_{\text{COV}} + y) \ ?$$

The problem PPSCOVER is the subproblem of SCOVER in which the given net A is additionally required to be a PP-net.

Proposition 6.4 (**EXPSPACE-hard** COVER **reduces to** SCOVER)**.**
The COVER *problem (see Proposition 6.3) is polynomially reducible to* SCOVER*.*

Proof. Consider an instance of COVER, that is, an ordinary strongly reversible d-dim net A and INIT, COV $\in [1,d]$. We extend A to a $2d$-dim net A', where the components $d+1, d+2, \ldots, 2d$ are referred to as the *store places*. These places allow for the removal of any potential imbalance that occurs when an action a produces more (or fewer) tokens than it consumes. Specifically, $A' \subseteq \{0,1\}^{2d} \times \{0,1\}^{2d}$ is constructed as follows:

- Each action $a = (a_-, a_+) \in \{0,1\}^d \times \{0,1\}^d$ in A gives rise to
  the action $a' = (a'_-, a'_+) \in \{0,1\}^{2d} \times \{0,1\}^{2d}$ in A', where
  - $a'_-(i) = a_-(i)$ and $a'_+(i) = a_+(i)$ for all $i \in [1,d]$;
  - if $k = ||a_-||_1 - ||a_+||_1 \geq 0$, then
    - $a'_-(i) = 0$ for all $i \in [d+1, 2d]$,
    - $a'_+(i) = 1$ for all $i \in [d+1, d+k]$,
    - $a'_+(i) = 0$ for all $i \in [d+k+1, 2d]$;
  - if $k = ||a_+||_1 - ||a_-||_1 > 0$, then
    - $a'_-(i) = 1$ for all $i \in [d+1, d+k]$,
    - $a'_-(i) = 0$ for all $i \in [d+k+1, 2d]$,
    - $a'_+(i) = 0$ for all $i \in [d+1, 2d]$.

  (Hence, $||a'_-||_1 = ||a'_+||_1$.)
- Moreover, A' contains actions $(\mathbf{e}_{d+1}, \mathbf{e}_{d+d})$ and $(\mathbf{e}_{d+d}, \mathbf{e}_{d+1})$, and for each $i \in [1, d-1]$ it contains actions $(\mathbf{e}_{d+i}, \mathbf{e}_{d+i+1})$ and $(\mathbf{e}_{d+i+1}, \mathbf{e}_{d+i})$. (Hence, any token in a store place can move freely to any other store place.)

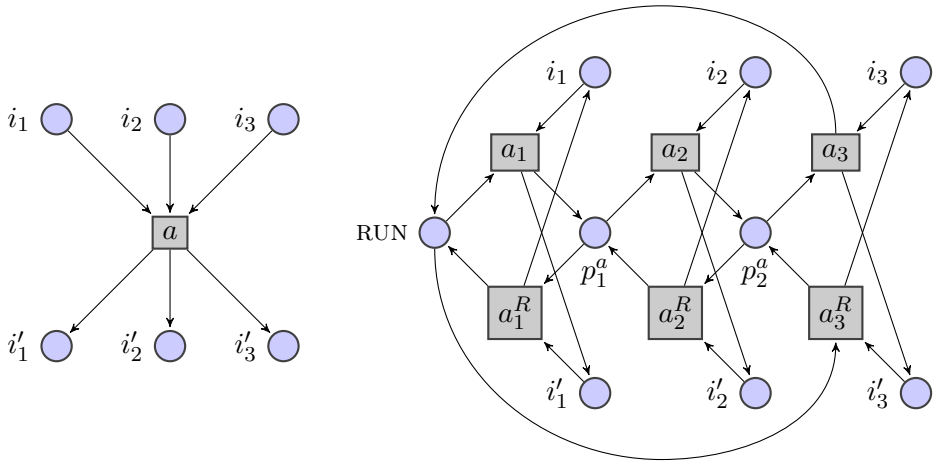


FIGURE 4. Transformation of the pair (a_-, a_+), (a_+, a_-) where $||a_-||_1 = ||a_+||_1 = 3$.

Since A is ordinary and strongly reversible, it is clear that A' is also ordinary and strongly reversible; moreover, A' is 1-conservative. It is easily verified that $(A, \text{INIT}, \text{COV})$ is a positive instance of COVER if and only if $(A', \text{INIT}, \text{COV}, d+1)$ is a positive instance of SCOVER. Indeed, if COV is coverable from INIT by an execution of A, then it suffices to provide an adequate number of tokens in the store place $d+1$ to allow the execution to be mimicked by A'. Conversely, the projection of any execution of A' onto the first d components constitutes a valid execution of A. □

Reducing SCOVER to PPSCOVER. We now aim to reduce the SCOVER problem to its subproblem PPSCOVER, which requires the input net to be a PP-net. This reduction is based on the following construction, which is an instance of the standard approach of simulating an action by a sequence of simpler actions, controlled by additional *control places*.

For convenience, an action $a = (a_-, a_+)$ of an ordinary 1-conservative d-dim net A with $||a_-||_1 = ||a_+||_1 = k$ is also written as

$$a : (i_1, i_2, \ldots, i_k) \to (i'_1, i'_2, \ldots, i'_k), \text{ or simply } (i_1, i_2, \ldots, i_k) \to (i'_1, i'_2, \ldots, i'_k),$$

where $\{i_j \mid j \in [1,k]\} = \{i \mid a_-(i) = 1\}$ and $\{i'_j \mid j \in [1,k]\} = \{i' \mid a_+(i') = 1\}$. When $k = 1$, we write $a : i_1 \to i'_1$. Furthermore, recall that for a d-dim net A, the set $P = \{1, 2, \ldots, d\}$ is viewed as the set of *places* of A.

Proposition 6.5 (**1-conservative nets simulated by PP-nets**).
There exists a polynomial-time construction that transforms any ordinary strongly reversible 1*-conservative net A with a set of places P into an ordinary strongly reversible PP-net A' with a set of places $P' \supseteq P$ and a distinguished place* RUN $\in P' \setminus P$ *satisfying the following two conditions:*

(1) For all markings x, y of A: $x \xrightarrow{*} y$ for A if and only if $x' \xrightarrow{*} y'$ for A', where x' and y' are obtained from x and y, respectively, by adding one token to the place RUN *(and no tokens to the places in $P' \setminus (P \cup \{\text{RUN}\})$).*

(2) For each marking x of A': if $x_{|P' \setminus P} = \mathbf{0}$, then no action is enabled at x; if $||x_{|P' \setminus P}||_1 = 1$, then $||y_{|P' \setminus P}||_1 = 1$ for all $y \in \text{Reach}(x)$.

Proof. Let A be an ordinary strongly reversible 1-conservative net with a set of places P. To extend A to the required PP-net A', we introduce a new control place RUN, together with further auxiliary places as specified by the following construction of the actions of A'.

For each pair of mutually reversed actions $a = (a_-, a_+)$ and $a^R = (a_+, a_-)$ in A, written as

$$a : (i_1, i_2, \ldots, i_k) \to (i'_1, i'_2, \ldots, i'_k) \text{ and } a^R : (i'_1, i'_2, \ldots, i'_k) \to (i_1, i_2, \ldots, i_k),$$

we proceed as follows (see Figure 4):

(a) If $k = 1$, and we have $a : i_1 \to i'_1$ and $a^R : i'_1 \to i_1$, then A' contains the (mutually reversed) actions
$$a_1 : (i_1, \text{RUN}) \to (i'_1, \text{RUN}) \text{ and } a_1^R : (i'_1, \text{RUN}) \to (i_1, \text{RUN}).$$

(b) If $k > 1$ then A' contains fresh auxiliary places $p^a_1, p^a_2, \ldots, p^a_{k-1}$, and the following actions:
  (i) $a_1 : (i_1, \text{RUN}) \to (i'_1, p^a_1)$ and $a_1^R : (i'_1, p^a_1) \to (i_1, \text{RUN})$;
  (ii) $a_j : (i_j, p^a_{j-1}) \to (i'_j, p^a_j)$ and $a_j^R : (i'_j, p^a_j) \to (i_j, p^a_{j-1})$, for all $j \in [2, k-1]$,
  (iii) $a_k : (i_k, p^a_{k-1}) \to (i'_k, \text{RUN})$ and $a_k^R : (i'_k, \text{RUN}) \to (i_k, p^a_{k-1})$.

The net A' is clearly an ordinary strongly reversible PP-net; the definition of its actions ensures that condition (2) holds. A step $x \xrightarrow{a} y$ of A is simulated by the execution $x' \xrightarrow{a_1 \cdots a_k} y'$ of A', while $y \xrightarrow{a^R} x$ is simulated by $y' \xrightarrow{a_k^R \cdots a_1^R} x'$, where x' and y' are obtained from x and y by placing a token in RUN. Condition (1) thus holds as well. □

Proposition 6.6 (SCOVER **reduces to** PPSCOVER)**.**
The problem SCOVER *is polynomially reducible to* PPSCOVER*.*

Proof. Let $(A, \text{INIT}, \text{COV}, \text{STORE})$ be an instance of SCOVER, where A is an ordinary strongly reversible 1-conservative net with a set of places P. Let A' be the PP-net obtained from A via Proposition 6.5, with a set of places $P' \supseteq P$, where $P' \setminus P$ contains the distinguished place RUN. We construct A'' from A' by adding two new places INIT′ and COV′; each action $a = (a_-, a_+) \in A'$ is extended to an action in A'' by setting $a_-(i) = a_+(i) = 0$ for $i \in \{\text{INIT}', \text{COV}'\}$.

Moreover, let A'' also contain the following actions, each together with its reverse:

$$(\text{INIT}', \text{STORE}) \to (\text{INIT}, \text{RUN}),\ (\text{COV}, \text{RUN}) \to (\text{COV}', \text{STORE}),$$
$$(\text{INIT}, \text{RUN}) \to (\text{INIT}', \text{STORE}),\ (\text{COV}', \text{STORE}) \to (\text{COV}, \text{RUN}).$$

Thus, A'' is an ordinary strongly reversible PP-net. We conclude the proof by verifying that $(A, \text{INIT}, \text{COV}, \text{STORE})$ is a positive instance of SCOVER iff $(A'', \text{INIT}', \text{COV}', \text{STORE})$ is a positive instance of PPSCOVER.

($\Rightarrow$): Let Exec be an execution of A demonstrating $(\mathbf{e}_{\text{INIT}} + k \cdot \mathbf{e}_{\text{STORE}}) \xrightarrow{*} (\mathbf{e}_{\text{COV}} + y)$. By Proposition 6.5(1), there exists a corresponding execution Exec′ of the PP-net A' where both the initial marking $\mathbf{e}_{\text{INIT}} + k \cdot \mathbf{e}_{\text{STORE}}$ and the target marking $\mathbf{e}_{\text{COV}} + y$ are augmented by a token in RUN. Consequently, if A'' starts at $\mathbf{e}_{\text{INIT}'} + (k{+}1) \cdot \mathbf{e}_{\text{STORE}}$, it can perform the action $(\text{INIT}', \text{STORE}) \to (\text{INIT}, \text{RUN})$, simulate the execution Exec′ of A', and conclude by performing $(\text{COV}, \text{RUN}) \to (\text{COV}', \text{STORE})$.

($\Leftarrow$): Let Exec be an execution of A'' demonstrating $(\mathbf{e}_{\text{INIT}'} + k \cdot \mathbf{e}_{\text{STORE}}) \xrightarrow{*} (\mathbf{e}_{\text{COV}'} + y)$. By Proposition 6.5(2), we deduce that Exec maintains exactly one token within the set of places $\{\text{INIT}', \text{COV}'\} \cup (P' \setminus P)$. Furthermore, whenever the token resides in INIT′, the only enabled action is $(\text{INIT}', \text{STORE}) \to (\text{INIT}, \text{RUN})$. Consequently, Exec can be (re)constructed so that it starts with $(\text{INIT}', \text{STORE}) \to (\text{INIT}, \text{RUN})$, simulates an execution Exec′

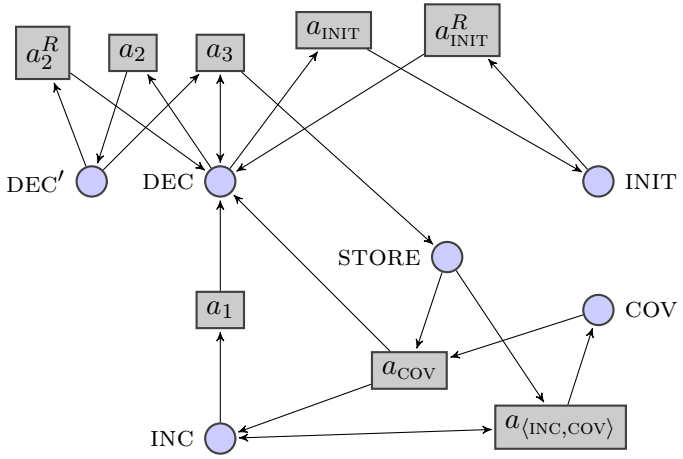


FIGURE 5. Construction in the proof of Lemma 6.7.

of A', and concludes with $(\text{COV}, \text{RUN}) \to (\text{COV}', \text{STORE})$. By Proposition 6.5(1), the existence of Exec' implies that there is an execution of A starting from $(\mathbf{e}_{\text{INIT}} + (k{-}1) \cdot \mathbf{e}_{\text{STORE}})$ that reaches a marking with a token in COV. □

The proof of Theorem 6.1 is thus completed by the following lemma. Note that this reduction does not preserve *strong* reversibility.

Lemma 6.7 (PPSCOVER **reduces to structural liveness**)**.**
The PPSCOVER *problem is polynomially reducible to the structural liveness problem for ordinary reversible PP-nets.*

Proof. Let $(A, \text{INIT}, \text{COV}, \text{STORE})$ be an instance of PPSCOVER. Thus, A is an ordinary strongly reversible d-dim PP-net, with the set P of d places, three of which are INIT, COV, STORE. We extend A to an ordinary, reversible (though not strongly reversible) PP-net A' such that $(A, \text{INIT}, \text{COV}, \text{STORE})$ is a positive instance of PPSCOVER if and only if A' is structurally live.

The set P' of places of A' is constructed from P by adding fresh places $\text{INC}, \text{DEC}, \text{DEC}'$; hence A' is a $(d{+}3)$-dim net, and each $a = (a_-, a_+) \in A$ is extended to become an action in A', by setting $a_-(i) = a_+(i) = 0$ for all $i \in \{\text{INC}, \text{DEC}, \text{DEC}'\}$. The net A' further contains the following actions, partially depicted in Figure 5:

(1) $a_{\text{COV}} : (\text{COV}, \text{STORE}) \to (\text{INC}, \text{DEC})$;
(2) for each $p \in P \setminus \{\text{STORE}\}$,
$$a_{\langle \text{INC},p\rangle} : (\text{INC}, \text{STORE}) \to (\text{INC}, p) \text{ and } a_{\langle \text{DEC},p\rangle} : (\text{DEC}, p) \to (\text{DEC}, \text{STORE});$$
(3) $a_1 : \text{INC} \to \text{DEC}$, $a_2 : \text{DEC} \to \text{DEC}'$; $a_2^R : \text{DEC}' \to \text{DEC}$,
(4) $a_3 : (\text{DEC}, \text{DEC}') \to (\text{DEC}, \text{STORE})$;
(5) $a_{\text{INIT}} : \text{DEC} \to \text{INIT}$ and $a_{\text{INIT}}^R : \text{INIT} \to \text{DEC}$.

Since A is an ordinary strongly reversible PP-net, it is evident that A' is an ordinary PP-net. We now show that A' is also (structurally) reversible: Since a^R_{INIT} is the reverse of a_{INIT}, a^R_2 is the reverse of a_2, and $\Delta(a_{\langle \text{INC},p\rangle}) = -\Delta(a_{\langle \text{DEC},p\rangle})$ (for each $p \in P \smallsetminus \{\text{STORE}\}$), it suffices to show a sequence σ of actions of A' such that $\Delta(a_{\text{COV}}\, a_1\, a_3\, \sigma) = \mathbf{0}$, i.e., $\Delta(\sigma)(\text{DEC}) = -2$, $\Delta(\sigma)(\text{DEC}') = 1$, $\Delta(\sigma)(\text{COV}) = 1$, and $\Delta(\sigma)(p) = 0$ for all other places p. We can take $\sigma = a_2\, a_2\, a_3\, a_{\langle \text{INC},\text{COV}\rangle}$.

We call a *marking* x of A' *initial* if $x(\text{INIT}) = 1$ and $x(p) = 0$ for all places p outside the set $\{\text{INIT}, \text{STORE}\}$. We further identify the following "initial" *property* IP: for every marking x of A' with $x(p) \geq 1$ for some $p \in P' \smallsetminus P = \{\text{INC}, \text{DEC}, \text{DEC}'\}$ there is an initial marking x' such that $x \xrightarrow{*} x'$ (which entails $||x'||_1 = ||x||_1$ since A' is conservative, which in turn makes this initial marking unique). Indeed, it suffices to use the actions a_1, a_2, a^R_2, a_3, $a_{\langle \text{DEC},p\rangle}$ (for $p \in P \smallsetminus \{\text{STORE}\}$), and a_{INIT} in an appropriate sequence.

We now verify that $(A, \text{INIT}, \text{COV}, \text{STORE})$ is a positive instance of PPSCover if and only if A' is structurally live.

($\Leftarrow$): Let x_0 be a live marking of A'. Since a_{COV} is not dead at x_0, by IP we deduce that there is an initial marking x_1 such that $x_0 \xrightarrow{*} x_1$; x_1 is necessarily live. Let $x_1 \xrightarrow{\sigma} x_2$ be a shortest execution from x_1 enabling a_{COV}; hence $x_2(\text{COV}) \geq 1$. Since shortest, σ cannot contain any action from the set $\{a_{\text{COV}}, a_1\} \cup \{a_{\langle \text{INC},p\rangle} \mid p \in P \smallsetminus \{\text{STORE}\}\}$. Let ℓ be the number of occurrences of actions from the set $\{a^R_{\text{INIT}}, a_2, a^R_2, a_3, a_{\text{INIT}}\} \cup \{a_{\langle \text{DEC},p\rangle} \mid p \in P \smallsetminus \{\text{STORE}\}\}$ in σ. Let σ' arise from σ by omitting all these occurrences, and let x'_1 arise from x_1 by increasing the number of tokens in STORE by ℓ (to be generous). Then we clearly have $x'_1 \xrightarrow{\sigma'} x'_2$, where $x'_2(\text{COV}) \geq 1$; since $(x'_1)_{|P} \xrightarrow{(\sigma')_{|P}} (x'_2)_{|P}$ is an execution of A, we get that $(A, \text{INIT}, \text{COV}, \text{STORE})$ is a positive instance of PPSCover.

($\Rightarrow$): Let k, y be such that $(\mathbf{e}_{\text{INIT}} + k \cdot \mathbf{e}_{\text{STORE}}) \xrightarrow{*} (\mathbf{e}_{\text{COV}} + y)$ in A and, moreover, $k \geq |P|$. We show that the initial marking x_0 of A' for which $x_0(\text{STORE}) = k + 1$ (and $x_0(\text{INIT}) = 1$) is live.

Note that no action is dead at x_0: we can enable and execute a_{COV} that puts a token in both INC and DEC, and by using the actions $a_{\langle \text{INC},p\rangle}$ and $a_{\langle \text{DEC},p\rangle}$ we can then enable any action, except a^R_2 and a_3; but executing the enabled a_2 enables a^R_2, and a_1 then enables a_3.

To complete the proof, it suffices to show that x_0 is a home-marking, that is, $x_0 \xrightarrow{*} x$ implies $x \xrightarrow{*} x_0$. For the sake of contradiction, suppose that $x_0 \xrightarrow{*} x_1 \xrightarrow{a} x_2$ is an execution where $x_1 \xrightarrow{*} x_0$ and $x_2 \not\xrightarrow{*} x_0$. This implies that a has no reverse action, and it is thus an action among $a_{\text{COV}}, a_1, a_3, a_{\langle \text{INC},p\rangle}, a_{\langle \text{DEC},p\rangle}$. But then $x_2(\text{DEC}) \geq 1$ or $x_2(\text{INC}) \geq 1$, and by IP we get $x_2 \xrightarrow{*} x_3$ for an initial marking x_3. Since $||x_3||_1 = ||x_0||_1$, we get $x_3 = x_0$, which contradicts $x_2 \not\xrightarrow{*} x_0$. □

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